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Dirichlet Process Mixture Models for Identifying Independent non-Gaussian Shocks in SVARs: An Application to Global Crude Oil Market and Energy Production

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Chapter 1

Introduction

1.1 Macroeconomic background

In the 1960s and 1970s, there was a debate in the USA about the causality of monetary aggregates to national income. Based on historical data from the US economy, Friedman and Schwartz (1963 [1] pp. 676-680) showed that money growth (through monetary policy) and income growth were positively correlated and concluded that a constant money growth rule would stabilize the business cycle. This proposal evolved into the so-called „monetarism", which at that time appeared as an argument against Keyniasism with the perception that monetary policy supposedly would explain economic fluctuations much better than fiscal policy.[1] Some Keynesian economists challenged the effectiveness of a strict monetary control policy later; e.g., Tobin (1970 [2] pp. 312-314) used the money multiplier (which is not fixed) of the banking system as a great example to show that it is rather changes in real economic activity that provoke changes in monetary aggregates and not the contrary. At first glance, the causality test introduced by Granger (1969 [3] pp. 428-430) seemed to prove in favor of monetarists. Sims (1972 [4] pp. 540-542), however, pointed out serious limitations of Granger tests for macroeconomic data such as omitted variables and simultaneity biases, the lack of enough structural causality when applied to monetary economics, and later (1980 [5] pp. 1-4 & 16-20) even proved its inconsistency for the US case.[2]

The debate evolved to mutual recognition that the main interest is not the correlation between money and income growth, nor to test a particular lead-lag pattern, but rather to explain

[1] In monetarism more weight is putted on money supply to explain changes in economic activity, and even inflation in the long run. Traditional Keynesians assumed that fiscal policy was the main driver of economic fluctuations. New Keyniasism considers that both monetary and fiscal policy matter, but fiscal policy is more important in crises.

[2] According to Sims, Granger tests are very sensible to additional independent variables and some statistical properties. In particular, he showed that in a Granger framework, income changes might also cause money supply movements, implying that central banks respond to economic conditions rather than unilaterally driving them. To correct this contradiction, he introduces the Vector Autoregression (VAR) setup, which allows for a multivariate approach and reduces the risk of misspecification. Lütkepohl (1982 [6] pp. 372–375), later applied the VAR framework to show that the results of the Granger test extensively reflect variable omission.

how income responds to unanticipated changes in money growth (innovations or shocks). In the late 1980s, the structural VAR (SVAR) model proved to be the most suitable lever for identifying exogenous shocks, making it possible to distinguish between the sources of innovations in monetary policy, demand, supply, to plan unanticipated changes, recover shocks from previous expectations, and quantify cause-and-effect relationships based on clear economic restrictions.[3]

As a consequence of the debate between monetarists and Keynesians on the clarification of causal effects with respect to contemporaneous economic variables, the question arises of how to describe the effects of the crude oil price on stock prices. In the 1980s, it was already clear that crude oil production has a strong impact on the real price of oil, which in turn significantly accounts for fluctuations in stock aggregates.[4] The switch from VAR to SVAR models and the oil price boom of the 2000s have increased interest in structural analysis and led to the differentiation between supply-driven and demand-driven oil price innovations. To explain these effects on stock markets, Kilian (2009 [11] pp. 1055–1057) first proposed an SVAR model of the global market for crude oil. It considers the n-dimensional vector of endogenous variables

$$y_t = (\Delta Q_t, X_t, P_t)^{'}, \tag{1.1}$$

where ΔQ_t stands for the percent change of the world crude oil production (supply), X_t a business cycle index measuring global real economic activity (a proxy for oil demand such as industrial production or GDP), P_t the log of the real price of oil, and the vector of monthly identified reduced-form errors is given as follows:

$$\begin{pmatrix} u_t^{\Delta Q} \\ u_t^{X} \\ u_t^{P} \end{pmatrix} = \begin{bmatrix} b_0^{11} & 0 & 0 \\ b_0^{21} & b_0^{22} & 0 \\ b_0^{31} & b_0^{32} & b_0^{33} \end{bmatrix} \begin{pmatrix} \xi_t^{\text{oil supply}} \\ \xi_t^{\text{oil-specific demand}} \\ \xi_t^{\text{aggregate demand}} \end{pmatrix}. \tag{1.2}$$

The b_0^{ij} represent the weights attached to the structural shocks $\xi_t^{\bullet}$. The triangular structure of the weight matrix reflect recursive identifying restrictions implied by the particular economic situation. Here, for example, the first equation $u_t^{\Delta Q} = b_0^{11}\xi_t^{\text{oil supply}} + 0 + 0$ implies that the production of crude oil is only affected by supply-driven shocks, w.l.o.g. $b_0^{11} \neq 0$. The price of oil is influenced by all three shocks simultaneously. In an orthonormal reference system, the change in quantity on the abscissa and the change in price on the ordinates, the demand and supply curves are not necessarily orthogonal. Conditional on lags of all variables, the configuration of the weights in equation (1.2) exhibits a vertical short-term oil supply and a downward sloping short-term oil demand curve.

[3] A formal presentation of the Structural Vector Autoregression (SVAR) model appears in Sims (1986 [7] pp. 2-5). He builds on the 1980 VAR methodology to explicitly define the SVAR framework, discusses how economic theory can be used to impose identification restrictions on VAR residuals, and shows how to use the Cholesky decomposition for exogenous recursive shock identification. A refinement of his work around 1993 [8] contains evidence that structural shocks are better identified using probabilistic models.

[4] See, e.g., Hamilton (1983 [9] p. 229) and Chen, Roll & Ross (1986 [10] p. 401)

1.2 SVARs and the challenges with non-Gaussian shocks

Kilian (2009 [11] pp. 1064-1066) built historical decompositions from the late 1970s that establish oil price fluctuations to be primarily demand-driven. Kilian and Murphy (2014 [12]), Zhou (2020 [13]), Cross, Nguyen, and Tran (2022 [14]) arrived at the same results. However, Baumeister and Hamilton (2019 [15]), Caldara, Cavallo, and Iacoviello (2019 [16]) reported the opposite, i.e., oil prices would be primarily supply driven and demand shocks would be rather negligible. Braun (2023 [17]) attributes the divergence in the results to the different identification strategies used. His opinion is based on the fact that when a very low upper limit is imposed on the short-term price elasticity of supply, it becomes natural to obtain also minor supply shock effects, and the same reasoning is true for the price elasticity of demand. Analogously to Braun (2023 [17]), Fiorentini and Sentana (2023 [18]) demonstrate that the specification of a wrong marginal distribution increases the probability of inconsistent shock estimates and of respective standard deviations, which, in turn, invalidates the inference on the variance decomposition of the forecast error. Simply put, price elasticities and the distribution implied by the generating process of the model errors are crucial for attributing right shock effects consistently.

Classical SVAR models often assume a Gaussian distribution for structural shocks when trying to identify and estimate them. However, a great number of real-world economic shocks frequently embed non-Gaussian tendencies or characteristics, especially for financial concerns like oil prices, which often are very unstable. In times of recession, one is most likely faced with heavy tails, skewness, or multimodal distributions. If a Gaussianity test reveals non-Gaussianity, for example, if the null hypothesis of the Jarque-Bera test is rejected, the SVAR estimation process must absolutely be adapted to this end. On the other hand, if true independence of the shocks is additionally required instead of only uncorrelatedness, then a standard SVAR identification method such as the Cholesky decomposition becomes biased and leads therefore to false impulse response functions.[5] The current work shows how combining SVAR with DPMMs can efficiently resolve identification by both independence and non-Gaussianity simultaneously, and at the same time remain robust to outliers. Using Bayesian nonparametrics allows one to provide greater flexibility, avoid misleading policy conclusions, and reflect reliable statistical inference on macroeconomic variables, time-varying structures, and nonlinear relationships. Other popular alternatives to address identification by independence and non-Gaussianity such as ICA (Independent Component Analysis), higher-order moments such as skewness or kurtosis, are unfortunately often not robust to outliers. GARCH-SVARs are rather suitable for modeling volatility clustering and conditional heteroskedasticity. They are not necessarily the best choice for capturing nonparametric heterogeneity and are not as flexible as DPMM-SVARs to handle structural relationships in a flexible way.

[5]Although the Cholesky decomposition ensure shocks orthogonality, it account for independent shocks only if they are Gaussian. The triangular structure of the weights implies a recursive ordering instead of inherent statistical independence.

1.3 Motivation, relevancy and objectives

An important added value in the SVAR analysis is making use of very flexible estimation schemes to intuitively assign economic meanings to the identified shocks. Unfortunately, the majority of existing methods have some drawbacks that cannot be neglected, as they embed statistical properties that may not reflect contemporaneous economic restrictions. Using only non-Gaussianity, for example, in general yields a collection of independent shocks, which intrinsically are not useful for economic analysis. To make the interpretation of the shocks spontaneous and clear at all stages of the analysis, it is necessary to use an approach where non-Gaussianity is only utilized with consequent economic restrictions, probably precisioned by an adequate mix of prior distributions. In addition, the produced system of equations must be unique and order-invariant. The assumption of both non-Gaussian and independent shocks synchronously is unavoidable. Moreover, the estimation of the shocks are optimally reached with the help of flexible Bayesian models, as shown in Sims (1993 [8]). The advantages thus are numerous. First, Bayesian priors allow parameters to easily be shrunk toward economically meaningful values, helping to stabilize estimations. In this way, excessive variability is prevented in forecasts, especially in models with short time series and a lot of variables. Second, Bayesian SVAR estimation allows for uncertainty quantification, making forecasts more informative. Thus, by generating probability distributions over future outcomes rather than just point estimates, policy- and decision-makers can control margins much better, e.g. by means of credible intervals around predictions, and so allow one to assess the amount of non-Gaussianity in the data. Lastly, the ordering problem from, e.g. the Cholesky decomposition is avoided when interpreting shocks, since Bayesian methods can allow for weaker, more flexible restrictions. The central aim is to find a general framework in which estimation, identification, and implication of the shocks remain robust in the widest set of constellations, and can predictably easily be fitted to potential new arose restrictions, similar to the one of Braun (2023 [17]).[6]

1.4 Strategy and structure

Key pioneering works on DPMMs return to Ferguson (1973 [21]) who introduced the Dirichlet process (DP) as a distribution over distributions and Antoniak (1974 [22]) who generalized this approach to DPMMs (infinite mixture model with DP prior). Escobar & West (1995 [19]) developed Bayesian inference methods to estimate the concentration parameter (α) consistently in a hierarchical DPMM using MCMC. Neal (2000 [23]) made DPMM algorithms faster and scalable through slice sampling and variational inference. As an extension of identification by ICA, which

[6]Braun (2023 [17]) base his analysis on several algorithms, among other the one of Escobar and West (1995 [19]) for DPMMs. Besides frequentist kernel density estimation (KDE), Bayesian DPMMs account for renowned nonparametric methods to estimate the marginal distribution of structural shocks left unspecified. Braun (2023 [17]) also provides evidence in Table 4, that the posterior confidence in the oil market model is more narrow under the DPMM than under a parametric setup that relies on student-t distributions. Find a robust kernel approach to estimate SVARs in Hafner, Herwartz, and Wang (2024 [20]).

rely on fixed functional forms of the prior distribution relative to a Gaussian distribution, DPMMs ranks as arbitrary non-Gaussian shock estimation, whereas the number of components in the mixture is inferred without requiring explicit model selection criteria like AIC or BIC. The current work goes step by step through the application of current advancements in relation with DPMMs to estimate macroeconomic shocks expressed in a SVAR framework. First, in order to provide the reader with an intuitive representation of the identification scheme, the setup of the SVAR model and the general estimation of Bayesian parameters are described. Thereby, structural tools, namely the calculation of impulse responses (IRFs) and forecast error variance decompositions (FEVDs), are briefly discussed. To build the Bayesian methodology on the right mathematical ground, the theoretical aspects of DPMMs and their integration in SVARs are then considered. The following part focuses on Bayesian inference. There, the full MCMC algorithm is revealed, convergence properties, and particular consequences for marginal likelihoods are derived. The fourth section is consecrated to applying the previous theory to characterize the impact of demand and supply on oil prices, statistically. The data is a collection of significant global crude oil market representatives from January 1958 to February 2022, which are sampled monthly, partly from `https://www.eia.gov/totalenergy/data/monthly/`. The SVAR-DPMM mechanism by which the shocks are estimated and resulting IRFs indicate a minor impact of supply-driven shocks on oil prices, for the short-term price elasticity of supply end up being concentrated near zero. Lastly, the implications for the formulation of policies and regulations with regard to CO_2 emissions and natural energy sources are substantially explained.

Chapter 2

Estimation of SVARs with non-Gaussian shocks

2.1 Model setup

The SVAR approach is based on breaking down data variations that cannot be predicted based on antecedent data into reciprocally uncorrelated exogenous shocks with a clear economic interpretation. Since shocks are considered latent, they must not be directly observable. An SVAR is designed to embed K model variables that are propelled by K distinct shocks of which the respective generating processes are approximated. Let y_t, $t = 1, ..., T$ be a K-dimensional time series and assume that the data generating process for y_t is optimally approximated by a VAR of order $p < \infty$. The structural VAR(p) model is given by

$$B_0 y_t = B_1 y_{t-1} + \cdots + B_p y_{t-p} + \xi_t = BY_{t-1} + \xi_t, \tag{2.1}$$

where the model coefficients B_i are $K \times K$ non-singular matrices, $i = 0, ..., p$, each y_t a $K \times 1$ vector with $\mathbb{E}(y_t) = 0$ for all $t = 1, ..., T$ (for example the mean-centered values), and the $K \times 1$ (latent) shocks ξ_t are mutually independent, uncorrelated, and $\xi_t \sim (0, \Sigma_\xi)$. Thus, $Y_{t-1}^{'} \equiv (y_{t-1}^{'}, ..., y_{t-p}^{'})^{'}$, respectively $B \equiv (B_1, \cdots, B_p)$. Deterministic terms can be added easily without perturbing any result in the following unfolding. Left multiplication by B_0^{-1} in equation (2.1) gives the reduced form

$$y_t = \underbrace{B_0^{-1} B_1}_{A_1} y_{t-1} + \cdots + \underbrace{B_0^{-1} B_p}_{A_p} y_{t-p} + \underbrace{B_0^{-1} \xi_t}_{u_t} = AY_{t-1} + u_t, \tag{2.2}$$

with the reduced-form weights $A_i = B_0^{-1} B_i$, $i = 1, ..., p$, $u_t = B_0^{-1} \xi_t \sim (0, \Sigma_u)$ is a white noise error term with positive definite covariance matrix $\Sigma_u = B_0^{-1} \Sigma_\xi B_0^{-1'}$, and $A \equiv (A_1, \cdots, A_p)$.

Another representation of equation (2.2) is

$$A(L)y_t = u_t,$$

where $A(L) = I_K - A_1 L - \cdots - A_p L^p$ is a matrix polynomial in the lag operator L, i.e. $Ly_t = y_{t-1}$, for all $t = 2, ..., T$. In the multivariate case where each y_t is a vector of more than one component as, e.g., in equation (1.1), the identification of structural shocks boils down to the estimation of the optimal weight matrix B_0. Thus, latent shocks are recovered from the observed reduced-form errors u_t through $\xi_t = B_0 u_t$. For the purpose of the following section, it is sometimes useful to assume in addition that $\Sigma_\xi = I_K$, such that $\Sigma_u = B_0^{-1} B_0^{-1'}$. Alternatively, note that statistical identification would require that diagonal elements of B_0 are standardized to one and the covariance of ξ_t is a diagonal matrix, i.e.

$$B_0 = \begin{bmatrix} 1 & b_{12} & \cdots & b_{1K} \\ b_{21} & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & b_{K-1,K} \\ b_{K1} & \cdots & b_{K,K-1} & 1 \end{bmatrix} \quad \text{and} \quad \Sigma_\xi = \begin{bmatrix} \sigma^2_{\xi_1} & & 0 \\ & \ddots & \vdots \\ 0 & \cdots & \sigma^2_{\xi_K} \end{bmatrix}.$$

2.1.1 Statistical identification by non-Gaussianity

The structure of the covariance matrix Σ_u is the main information by which identification is achieved successfully. Local identification is typically the best that can be achieved, meaning that the estimated structural parameters are uniquely determined only in a neighborhood of the true values, but may not be uniquely determined over the entire parameter space.[1] By assumption it is $\Sigma_\xi \equiv \mathbb{E}(\xi_t \xi_t^{'}) = I_K$, and because $u_t = B_0^{-1}\xi_t$, it holds

$$\Sigma_u \equiv \mathbb{E}(u_t u_t^{'}) = \mathbb{E}(B_0^{-1}\xi_t\xi_t^{'}B_0^{-1'}) = B_0^{-1}\underbrace{\mathbb{E}(\xi_t\xi_t^{'})}_{I_K}B_0^{-1'} = B_0^{-1}B_0^{-1'}. \tag{2.3}$$

$\Sigma_u = B_0^{-1}B_0^{-1'}$ can be seen as a system of nonlinear equations in the unknown parameters B_0^{-1}. Any square $K \times K$ matrix contains exactly K^2 elements. Because Σ_u is a symmetric matrix, it suggests a system of exactly $\sum_{i=0}^{K-1}(K-i) = \sum_{i=0}^{K-1} K - \sum_{i=1}^{K-1} i = K^2 - \frac{(K-1)K}{2} = K(K+1)/2$ independent equations. It can be solved for B_0^{-1} numerically, e.g., by the method of moments under the condition that the number of unknown parameters in B_0^{-1} does not exceed $K(K+1)/2$. Additional potential restrictions may reflect exclusion, proportionality, or equality. If $\tilde{B} := \mathcal{Q}^{'}\Sigma_\xi^{-1/2}B_0$ is considered for an orthogonal matrix $\mathcal{Q}$ (that is, $\mathcal{Q}\mathcal{Q}^{'} = I_K$), $\mathcal{Q}^{-1} = \mathcal{Q}$ and

[1] Note that all signs in a column of B_0 can always be reversed without changing the product $B_0^{-1}B_0^{-1'}$.

$\tilde{\Sigma}_\xi = I_K$, then the obtained covariance matrix is equivalent to the original. Thus, it holds

$$\tilde{B}^{-1}\tilde{\Sigma}_\xi\tilde{B}^{-1'} = B_0^{-1}\Sigma_\xi^{1/2}\mathcal{Q}\mathcal{Q}'\Sigma_\xi^{1/2}B_0^{-1'} = B_0^{-1}\underbrace{\Sigma_\xi}_{I_K}B_0^{-1'} = B_0^{-1}B_0^{-1'},$$

meaning that the covariance structure of the forecast errors jointly identifies B_0 and Σ_ξ up to a set of orthogonal rotation matrices summarized in $\mathcal{Q} = PD$ producing equivalent models, where P is a K-dimensional permutation matrix and D a diagonal matrix with elements $+1$ or -1. The empirical mutual independence of structural shocks can be tested, for example, using the Matteson and Tsay (2017 [24] sections 2.1 and 2.2) nonparametric test based on distance covariances. According to Manne, Meitz, and Saikkonen (2017 [25] pp. 289-291), the SVAR model is identified up to permutation, sign, and scale through the following assumptions:

1. ξ_t is strictly stationary and a random vector with $\mathbb{E}(\xi_{it}) = 0$ and $\mathbb{E}(\xi_{it}^2) < \infty$, $\forall i = 1, ..., K$.
2. $\xi_{it} \perp\!\!\!\perp \xi_{jt}, \forall\, 1 \leq i \neq j \leq K$, and at least $K-1$ structural shocks are non-Gaussian.
3. $\text{Cov}(\xi_{it}, \xi_{i,t-k}) = 0, \forall k \neq 0,\ i = 1, ..., K.$

Finally, consider the following small introductory example of the stylized bivariate model of supply and demand identified by the parameters γ (price elasticity of supply), λ (price elasticity of demand), σ_1 and σ_2 (standard deviations of supply and demand shocks respectively):

$$\begin{aligned} &Supply: && q_t = \gamma p_t + \sigma_1 \xi_t^s \\ &Demand: && q_t = \lambda p_t + \sigma_2 \xi_t^d. \end{aligned}$$

Here q_t represents changes in (log) quantities, p_t changes in (log) prices, and it is $\begin{pmatrix} \xi_t^s \\ \xi_t^d \end{pmatrix} \sim (0_2, I_2)$. Each shock can be expressed as a linear combination of respective quantities and prices:

$$\begin{aligned} &Supply: && \xi_t^s = \frac{q_t}{\sigma_1} - \frac{\gamma}{\sigma_1}p_t \\ &Demand: && \xi_t^d = \frac{q_t}{\sigma_2} - \frac{\lambda}{\sigma_2}p_t, \end{aligned}$$

or in matrix form

$$\underbrace{\begin{pmatrix} \xi_t^s \\ \xi_t^d \end{pmatrix}}_{\text{structural shocks } \xi_t} = \underbrace{\begin{bmatrix} \frac{1}{\sigma_1} & -\frac{\gamma}{\sigma_1} \\ \frac{1}{\sigma_2} & -\frac{\lambda}{\sigma_2} \end{bmatrix}}_{\text{weights } B_0} \underbrace{\begin{pmatrix} u_t^q \\ u_t^p \end{pmatrix}}_{\text{VAR errors } u_t}.$$

The shocks cannot be identified by the method of moments because the second moment of the data cannot be conclusive, as the number of structural parameters ($\sigma_1, \sigma_2, \gamma$ and λ) is not equal to the number of independent equations implied by the reduced-form error covariance matrix ($2(2+1)/2 = 3$). The standard way to challenge the estimation of the weight matrix B_0

is to impose additional restrictions on the structural parameters that correspond to economic theory, reflect prior economic knowledge, or any type of constraints on the structural model.[2] Small differences in the specification of these priors can have important consequences for the estimation of structural demand and supply shocks. Forecast errors in oil production and prices are quite uncorrelated, producing a great number of equally consistent sign-restricted models for just tiny changes. Because the model is not fundamentally structural if each ξ_t does not have a clear economic interpretation after identification, one must be very careful, as identified shocks by means only of the non-Gaussianity assumption would not be enough. To be sure that consequent shock identification is achieved with regard to the additional restrictions, it is major that the analysis leads to a joint posterior which reflects all economic identifying information at hand, implied simultaneously by the prior distribution and the likelihood.

2.1.2 Structural impulse response analysis

Now assume stationarity throughout, i.e.,

$$\det A(L) = \det(I_K - A_1 L - \cdots - A_p L^p) \neq 0 \quad \text{for } |L| \leq 1.$$

The inverse VAR operator is defined and given by $A(L)^{-1} = \sum_{i=0}^{\infty} \Phi_i L^i$, where the Φ_i's are obtained recursively by

$$\Phi_0 = I_K, \quad \text{and} \quad \Phi_i = \sum_{j=1}^{i} \Phi_{i-j} A_j, \qquad i = 1, 2, ..., \tag{2.4}$$

with $A_j = 0$ for $j > p$. There exist infinitely many MA representations $y_t = \eta + \sum_{i=0}^{\infty} \Phi_i u_{t-i}$, where $\eta = A(1)^{-1}\nu$ and ν is the deterministic term of the VAR(1) process. Any nonsingular linear transformation of the white noise process u_t, e.g. $\varepsilon_t := Q u_t$, produces another white noise process that can be used for another MA representation $y_t = \eta + \sum_{i=0}^{\infty} \Theta_i \varepsilon_{t-i}$, where $\Theta_i = \Phi_i Q^{-1}$, $i = 0, 1, ...$, contains the impulse response that captures the dynamic effect of transformed white noise ε_t on y_t, i periods after the shock.[3]

To understand how structural shocks affect the system over time, one typically uses impulse responses, where the unknown parameters A_i are replaced by consistent estimates in the reduced-form VAR(p) model. Having constructed consistent estimates $\widehat{A}_j$, $j = 1, ..., p$, $\widehat{\Sigma}_u$, and therefore having the implied estimate $\widehat{B}_0$ at hand, estimates $\widehat{\Phi}_i$ (and thus $\widehat{\Theta}_i$) can be constructed recursively according to equation (2.4). Considering the MA representation and looking out to a maximum propagation horizon H, the structural impulse response matrices Θ_i are deduced from

[2]The assumption on selected priors should reflect economic restrictions such as the sign and/or magnitude of the structural parameters. Conventional sign restrictions, for example, are a negative slope (λ) of demand and a positive slope (γ) of supply. However, in this work, the sign restrictions are not considered for the sake of simplicity. Find relative necessary adaptations in Braun and Brüggemann (2017 [26] sections 2.2 and 2.3)

[3]See Kilian and Lütkepohl (2017 [27]), section 2.2.2, *The Moving Average Representation*, prediction error, Wold MA representation.

the responses of each element of $y_t = (y_{1t}, ..., y_{Kt})^{'}$ to a single impulse in $\xi_t = (\xi_{1t}, ..., \xi_{Kt})^{'}$ after i period(s) as

$$\frac{\partial y_{t+i}}{\partial \xi_t^{'}} = \Theta_i, \quad \text{i= 0, 1, 2, ...,} \ H,$$

where Θ_i is a $K \times K$ matrix composed of the structural impulse responses

$$\theta_{jk,i} = \frac{\partial y_{j,t+i}}{\partial \xi_{kt}}$$

to a unit structural shock ξ_{kt} for given i so that $\Theta_i = [\theta_{jk,i}]$. There are K^2 impulse response functions in total, each of length $H + 1$. To get the reduced-form impulse responses Φ_i (or forecast errors) of the j-th variable to a unit VAR error u_{kt}, i periods ago, the VAR(1) representation of the VAR(p) process is used in combination with successive substitution for Y_{t-i}. This is, if

$$Y_t = AY_{t-1} + U_t,\ Y_t \equiv \begin{pmatrix} y_t \\ \vdots \\ y_{t-p+1} \end{pmatrix},\ A \equiv \begin{bmatrix} A_1 & A_2 & \dots & A_{p-1} & A_p \\ I_K & 0 & \dots & 0 & 0 \\ 0 & I_K & \dots & 0 & 0 \\ \vdots & & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & I_K & 0 \end{bmatrix},\ U_t \equiv \begin{pmatrix} u_t \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \text{ and } J \equiv$$

$[I_K, 0_{K\times K(p-1)}]$ is a row selector, then $Y_{t+i} = A^{i+1}Y_{t-1} + \sum_{j=0}^{i} A^j U_{t+i-j}$ implies

$$y_{t+i} = JA^{i+1}Y_{t-1} + \sum_{j=0}^{i} JA^j \underbrace{JJ^{'}}_{I_K} U_{t+i-j} = JA^{i+1}Y_{t-1} + \sum_{j=0}^{i} JA^j J^{'} \underbrace{JU_{t+i-j}}_{u_{t+i-j}}$$

where the substitution for $t - i$ yields $\underset{K\times K}{\Phi_i} = [\phi_{jk,i}] \equiv JA^iJ^{'}$ with respect to a unit shock u_{kt}, $k = 1, ..., K$. Φ_i can be considered a caliber for how distant a shock lies in the past ($\Phi_0 = I_K$). Using $\Theta_i \equiv \Phi_i B_0^{-1}$, the multivariate MA representation follows as

$$y_t = \sum_{i+0}^{\infty} \Phi_i u_{t-i} = \sum_{i+0}^{\infty} \Phi_i B_0^{-1} B_0 u_{t-i} = \sum_{i+0}^{\infty} \Theta_i \xi_{t-i}, \tag{2.5}$$

with the direct implication that

$$\frac{\partial y_t}{\partial \xi_{t-i}^{'}} = \frac{\partial y_{t+i}}{\partial \xi_t^{'}} = \Theta_i.$$

The Φ_i and Θ_i obtained in this way are consistent estimates for the coefficients of the structural MA representation. If the VAR is not stable, the impulse responses may not converge to zero for $i \to \infty$.

2.1.3 Forecast error variance decompositions (FEVDs)

To determine how much influence a structural shock ξ_{kt} has on the forecast error variance of y_{t+h} at some horizon $h = 0, ..., H$, one commonly considers the prediction error associated with an h-step ahead prediction, i.e.

$$y_{T+h} - y_{T+h|T} = u_{T+h} + \Phi_1 u_{T+h-1} + \cdots + \Phi_{h-1} u_{T+1}, \tag{2.6}$$

where $\Phi_i = JA^iJ'$. Replacing $\Phi_i u_{t+h-i}$ by $\Theta_i \xi_{t+h-i}$ as done before in equation (2.5) gives the h-step ahead forecast error

$$y_{t+h} - y_{t+h|t} = \sum_{i=0}^{h-1} \Phi_i u_{t+h-i} = \sum_{i=0}^{h-1} \Theta_i \xi_{t+h-i}.$$

At horizon h the mean-squared prediction error (MSPE) matrix is then given by

$$\text{MSPE}(h) \equiv \mathbb{E}[(y_{t+h} - y_{t+h|t})(y_{t+h} - y_{t+h|t})'] = \sum_{i=0}^{h-1} \Phi_i \Sigma_u \Phi_i' = \sum_{i=0}^{h-1} \Theta_i \underbrace{\Sigma_\xi}_{I_K} \Theta_i' = \sum_{i=0}^{h-1} \Theta_i \Theta_i'.$$

Because the model is stationary, the MSPE converges to the unconditional covariance matrix of y_t (i.e. $\Sigma_y(h) \underset{h\to\infty}{\to} \Sigma_y$) and the variance decomposition of y_t coincide with the limit of the FEVD as $h \to \infty$.[4] Only the Θ_i (computed for the structural impulse responses matrices) are needed to build the MSPE decomposition for horizon ∞. At horizon h, the contribution of the j-th shock to the MSPE of y_{kt} is given by

$$\text{MSPE}_j^k(h) = \theta_{kj,0}^2 + \cdots + \theta_{kj,h-1}^2,$$

where $\theta_{kj,h}$ denotes the kj-th element of Θ_h and the total MSPE of each y_{kt}, $k = 1, ..., K$ is

$$\text{MSPE}^k(h) = \sum_{j=1}^{K} \text{MSPE}_j^k(h) = \sum_{j=1}^{K} (\theta_{kj,0}^2 + \cdots + \theta_{kj,h-1}^2),$$

from which it follows

$$1 = \frac{\text{MSPE}^k(h)}{\text{MSPE}^k(h)} = \frac{\sum_{j=1}^{K} MSPE_j^k(h)}{\text{MSPE}^k(h)} = \frac{\text{MSPE}_1^k(h)}{\text{MSPE}^k(h)} + \frac{\text{MSPE}_2^k(h)}{\text{MSPE}^k(h)} + \cdots + \frac{\text{MSPE}_K^k(h)}{\text{MSPE}^k(h)},$$

for given h and k, and each ratio $\text{MSPE}_j^k(h)/\text{MSPE}^k(h)$ represent the fraction of the share of the j-th shock to the forecast error variance of variable k. The respective percentages are obtained by multiplication by 100.

[4] In integrated systems the MSPE diverges when the forecast horizon goes to infinity, but the FEVD remains valid up to a finite maximum horizon of H. In case of structural models with long-run restrictions, special methods,e.g., using the Beveridge-Nelson decomposition, can allow FEVD estimation. For more details, see, e.g. Kilian and Lütkepohl (2017 [27] section 2.1, p. 22), or Beveridge and Nelson (1981 [28]).

2.1.4 Bayesian SVAR estimation

Estimators derived from VAR models based on short samples usually tend to be imprecise when using standard frequentist approaches such as ML, LS, or the method of moments.[5] To reduce the high variances of the estimates, the analytical framework can be reinforced with estimates that shrink toward specific values. Bayesian approaches can also account for the inclusion of incidental or surprise-related economic information. In essence, frequentist methods would condition the estimation on the existence of a parameter, say $\boldsymbol{\vartheta}$, that controls the data generation process, the source of the observed sample. They would then infer the value of $\boldsymbol{\vartheta}$ taken as nonstochastic, neglecting the possibility of a stochastic data generation process, and thus always inferring probability statements based on repeated sampling properties in $\boldsymbol{\vartheta}$.

Bayesian estimation in contrast, treats the parameter vector $\boldsymbol{\vartheta}$ as stochastic and the data as nonstochastic, in the sense that the researcher's beliefs about the distribution of $\boldsymbol{\vartheta}$ prevail over the distribution of the data itself. As the Bayesian setup would eventually assume only a specific distribution (or family of distributions) for the data, it would not request a repeated sampling experiment, but rather a prior distribution for $\boldsymbol{\vartheta}$ reflecting subjective beliefs about its value before inspecting the data. The likelihood of the model in this form captures the combination of the researcher's prior beliefs about $\boldsymbol{\vartheta}$ with the distribution assumed for the data, to construct the posterior for $\boldsymbol{\vartheta}$. The posterior therefore reflects everything the researcher knows about the model parameter after having looked at the data.[6]

Let $g(\boldsymbol{\vartheta})$ be the prior probability density function summarizing the prior information about the distribution of $\boldsymbol{\vartheta}$, and $f(\boldsymbol{y}|\boldsymbol{\vartheta})$ the probability density function of the data conditional on $\boldsymbol{\vartheta}$.[7] Given the data $\boldsymbol{y} = (y_1^{'}, ..., y_T^{'})^{'}$, inference is conditional on the data $\boldsymbol{y}$. The Bayes theorem draws the joint distribution through

$$f(\boldsymbol{\vartheta},\ \boldsymbol{y}) = g(\boldsymbol{\vartheta}|\boldsymbol{y})f(\boldsymbol{y}), \tag{2.7}$$

from which it follows

$$g(\boldsymbol{\vartheta}|\boldsymbol{y}) = \frac{f(\boldsymbol{y},\ \boldsymbol{\vartheta})}{f(\boldsymbol{y})} = \frac{f(\boldsymbol{y}|\boldsymbol{\vartheta})g(\boldsymbol{\vartheta})}{f(\boldsymbol{y})},$$

where $g(\boldsymbol{\vartheta}|\boldsymbol{y})$ denotes the posterior probability density function and $f(\boldsymbol{y})$ is a normalizing constant representing the unconditional sample density, i.e., it holds in proportion

$$g(\boldsymbol{\vartheta}|\boldsymbol{y}) \propto f(\boldsymbol{y}|\boldsymbol{\vartheta})g(\boldsymbol{\vartheta}) = \ell(\boldsymbol{\vartheta}|\boldsymbol{y})g(\boldsymbol{\vartheta}). \tag{2.8}$$

[5]For some examples, see, e.g. Kilian (1998, [29] pp. 220–222), Sims (1980 [5] pp. 6-7), Kilian and Lütkepohl (1993 [27] pp. 218–220).

[6]Choosing between frequentist and Bayesian methodology relies on the particular case and preferences of the modeler. The results from both points of view must be separated. Even when the estimates are numerically the same, the interpretations do not follow the same logic. DPMMs may be seen as hybrid, for it is fundamentally Bayesian and nonparametric by infinite-dimensional DP priors, but it can sometimes estimate hyperparameters such as the concentration parameter (α) using frequentist methods.

[7]Algebraically, the conditional probability density function $f(\boldsymbol{y}|\boldsymbol{\vartheta})$ coincides with the likelihood function $\ell(\boldsymbol{\vartheta}|\boldsymbol{y})$.

Credible sets

A „credible set" is the term used in the Bayesian setup analogously to what a „confidence set" is to classical inference. Both are not constructed in the same manner and hence do not have the same interpretation. A $(1-\delta)100\%$ credible set for $\boldsymbol{\vartheta}$ with respect to the posterior $g(\boldsymbol{\vartheta}|\boldsymbol{y})$, $0<\delta<1$, is a set Ω such that

$$\mathbb{P}(\boldsymbol{\vartheta}\in\Omega|\boldsymbol{y})=\int_{\Omega}g(\boldsymbol{\vartheta}|\boldsymbol{y})d\boldsymbol{\vartheta}=1-\delta.$$

Sampling from the posterior distribution is generally not enough to estimate a unique credible set. A unique credible set is obtained if the highest posterior density set is chosen.[8] Because the posterior density is partly obtained from the prior, it is possible that the posterior for a particular parameter has more than one mode, leading to a highest posterior density area consisting in two or more disjoint sets. Even in the most hazard case, where the highest posterior density set and the frequentist confidence set are numerically the same, their interpretation is necessarily different. A frequentist $(1-\delta)100\%$ confidence set only ensures that the true value is included asymptotically (i.e., in repeated samples) with probability $1-\delta$, while a credible set gives a region where $(1-\delta)100\%$ of the probability mass is concentrated, facilitating probability statements about $\boldsymbol{\vartheta}$ based on a most probable stochastic model for the data $\boldsymbol{y}$. Building credible sets easily generalizes to smooth functions $h(\boldsymbol{\vartheta})$ like B-Splines, based on which the posterior can be approximated by simulation.

Simulating the posterior distribution with MCMC methods

To simulate the posterior distribution, random draws $\boldsymbol{\vartheta}^{(1)},...,\boldsymbol{\vartheta}^{(n)}$ are typically generated from the posterior of the parameters or some function of these parameters, such as smooth functions of $\boldsymbol{\vartheta}$, or the impulse responses associated with the VAR model. Suppose $h(\boldsymbol{\vartheta})$ denotes a vector or a scalar function, the expectation of which should be assessed asymptotically. By the law of large numbers, a good approximation is obtained when

$$\frac{1}{n}\sum_{i=1}^{n}h(\boldsymbol{\vartheta}^{(i)})\overset{a.s.}{\to}\mathbb{E}(h(\boldsymbol{\vartheta})|\boldsymbol{y})=\int h(\boldsymbol{\vartheta})g(\boldsymbol{\vartheta}|\boldsymbol{y})d\boldsymbol{\vartheta}, \tag{2.9}$$

where $\overset{a.s.}{\to}$ indicates the *almost sure* convergence, ensuring that eventually, with probability 1, the process would get stable at a certain limiting value. Of course the approximation precision increases with the number n of random draws. If the distribution in equation (2.9) is not standard or just unknown, Markov chains Monte Carlo (MCMC) is used in accordance with the central limit theorems for dependent samples to generate a sample of serially dependent values of $\boldsymbol{\vartheta}$ constructed such that its distribution converges to the posterior distribution of the parameters,

[8] That is Ω is chosen such that $g(\boldsymbol{\vartheta}|\boldsymbol{y})\geq c_\delta$ for all $\boldsymbol{\vartheta}\in\Omega$ and c_δ is the largest value with $\mathbb{P}(\boldsymbol{\vartheta}\in\Omega|\boldsymbol{y})=1-\delta$.

provided that the chain holds long enough.[9] It is a simple and efficient numerical method by which the posterior distribution can be estimated, instead of in closed form. Possible variations are for example *direct*, *acceptance* and *importance sampling* or *Gibbs samplers*. Differences revolve around only how the i-th draw $\boldsymbol{\vartheta}^{(i)}$ of $\boldsymbol{\vartheta}$ is chosen, given $\boldsymbol{\vartheta}^{(i-1)}$.

General Procedures for choosing the parameters of prior densities

An efficient idea to specify the prior as simple as possible is to define it such that the posterior belong to a known family of distributions. For example, when the prior is chosen such that the posterior has the same functional form as the prior, it is called a „conjugate prior". When the conjugate prior is from the same distribution family as the likelihood, it is called „natural conjugate prior ".[10] If the distribution of the prior densities can not be fully specified, one generally extends a given family of prior distributions by an additional structure with the goal to reduce the number of hyperparameters. As briefly mentioned in the last section, a narrow approximation to the posterior is most likely achieved when the burn-in (initialization) sample is as small as possible. Let $g(\boldsymbol{\vartheta}) = g_{\boldsymbol{\beta}}(\boldsymbol{\vartheta})$ where $\boldsymbol{\beta}$ denotes the vector of hyperparameters specifying the prior. The choice of $\boldsymbol{\beta}$ can rely on the forecast accuracy of the implied VAR model.[11] Regarding the prior as conditional on $\boldsymbol{\beta}$, it holds $g_{\beta}(\boldsymbol{\vartheta}) = g(\boldsymbol{\vartheta}|\boldsymbol{\beta})$, and a hierarchical prior is well adapted.[12] Using equation (2.8), the posterior is

$$g^*(\boldsymbol{\beta}) \propto h(\boldsymbol{y}|\boldsymbol{\beta})g(\boldsymbol{\beta}),$$

where the sample conditional density (i.e. the marginal likelihood) with respect to $\boldsymbol{\beta}$ is

$$h(\boldsymbol{y}|\boldsymbol{\beta}) = \int f(\boldsymbol{y}|\boldsymbol{\vartheta},\ \boldsymbol{\beta})g(\boldsymbol{\vartheta}|\boldsymbol{\beta})d\boldsymbol{\vartheta}. \tag{2.10}$$

The posterior of the hyperparameters is equal to the marginal likelihood if an improper (constant) uniform prior $g(\boldsymbol{\beta})$ is specified, in which case it makes sense to choose $\boldsymbol{\beta}$ such that $h(\boldsymbol{y}|\boldsymbol{\beta})$ is maximized. It will not integrate to one in a parameter space that is unbounded. However, it has the advantage that the marginal likelihood yield optimal out-of-sample forecasts and at the same time the procedure remain consistent from a frequentist point of view.[13]

[9] If a more precise inference is needed, one can simply consider only every m-th sampled vector where m is a sufficient large number, to construct approximately independent samples. A narrow approximation to the posterior is also facilitated by bounding the burn-in sample to a small number of initial sample values. For more material regarding the convergence diagnostic of the posterior to the target distribution, see, e.g, Chib (2001).

[10] An example of a conjugate prior is the normal prior for the VAR parameters for given Σ_u. The natural conjugate prior is the Gaussian-inverse Wishart prior. For details, see Kilian and Lütkepohl (2017 [27], sections 5.2.2 and 5.2.4).

[11] The choice of $\boldsymbol{\beta}$ may be based on out-of-sample forecasts, see, e.g. Doan, Litterman, and Sims (1984 [30] p. 11) or Litterman (1986 [31] p. 30), or based on the in-sample model fit, see, e.g. Banbura, Giannone, and Reichlin (2010 [32] p. 74).

[12] See, e.g., Koop (2012 [33] chapter 4, p. 126).

[13] See, e.g., Giannone, Lenza, and Primiceri ([34] pp. 439-441).

2.2 Inference using mixtures

Ferguson (1983 [35] pp. 293-295) discussed normal mixtures within the framework of DP priors, through which various problems of local versus global smoothing for uncertain distributions are addressed. Escobar and West (1995 [19]) proposed a more general computational method that enables evaluation of posterior and predictive distributions for all model parameters with the help of the MCMC method presented in Gelfand and Smith (1990 [36] section 2, p. 399). Let $(\Xi_t|\boldsymbol{\vartheta}_t) \sim \mathcal{N}(\mu_t, \sigma_t^2)$ be conditionally independent and normally distributed random variables from which the realizations of the structural shocks are supposed to originate, and where the parameter of interest $\boldsymbol{\vartheta}_t := (\mu_t, \sigma_t^2)$ comes from a prior distribution on $\mathbb{R} \times \mathbb{R}^+$, $t = 1, ..., T$, and the shocks $\mathcal{D}_T := \{\xi_1, ..., \xi_T\}$ are observed. That is, each ξ_t represents the observed value of Ξ_t. The distribution of a future case is a mixture of normals, and the density function $\Xi_{T+1} \sim \mathcal{N}(\mu_{T+1}, \sigma_{T+1}^2)$ is mixed with respect to the posterior predictive distribution for $(\boldsymbol{\vartheta}_{T+1}|\mathcal{D}_T)$. If the common prior distribution for $\boldsymbol{\vartheta}_t$ is uncertain and is modeled in whole or in part as a DP, then the shocks come from a Dirichlet mixture of normals.[14] Provided a suitable DP prior structure, predictive distributions are qualitatively similar to kernel techniques, plus different levels of smoothness are considered throughout the sample space, as different variances σ_t^2 are likely to be used. The posterior distribution combines information locally in the sample space to estimate the local structure, strongly supporting common values of individual parameters $\boldsymbol{\vartheta}_i$ and $\boldsymbol{\vartheta}_j$ for data points ξ_i and ξ_j that are close.

2.2.1 Dirichlet Processes (DPs)

A DP $G \sim \mathrm{DP}(G_0, \alpha)$ is a distribution over a distribution in the sense that it is random itself and uniquely characterized by two parameters, a base probability measure G_0 (location) and a concentration parameter $\alpha \in \mathbb{R}^+$ (scale). Each shock is assumed to be independent and to follow the hierarchical structure

$$\begin{aligned} \xi_t|\boldsymbol{\vartheta}_t &\sim F(\boldsymbol{\vartheta}_t), \\ \boldsymbol{\vartheta}_t &\sim G, \\ G &\sim DP(G_0, \alpha), \end{aligned}$$

where $F(\boldsymbol{\vartheta}_t)$ stands for a probability distribution parametrized by $\boldsymbol{\vartheta}_t$ with prior $\boldsymbol{\vartheta}_t \sim G$. The realization of a DP returns a.s. discrete priors for $\boldsymbol{\vartheta}_t$. On a measurable space $(\Omega, \mathcal{F}, G_0)$, $\mathcal{F}$ representing the sigma-algebra over all possible prior assumptions on $\boldsymbol{\vartheta}$, the $\mathrm{DP}(\alpha, G_0)$ is a random probability measure G over $(\Omega, \mathcal{F})$ such that for any finite measurable partition $(A_1, A_2, ..., A_r)$ of Ω, the random vector $(G(A_1), ..., G(A_r))$ is distributed as a finite-dimensional Dirichlet distribution

[14] See Escobar and West (1995 [19] p. 577) and Ferguson (1983 [35] p.287).

with parameters $(\alpha G_0(A_1), ..., \alpha G_0(A_r))$, i.e.,

$$(G(A_1), ..., G(A_r)) \sim \text{Dirichlet}(\alpha G_0(A_1), ..., \alpha G_0(A_r)).$$

The existence of the DP was established by Ferguson (1973 [21] pp. 210-212). If the mixing proportions $(G(A_1), ..., G(A_r))$ are given a symmetric Dirichlet prior with concentration parameter $\frac{\alpha}{r}$, a DPMM is obtained by letting $k \to \infty$. Representing a DPMM involves choosing between a marginal or conditional perspective for inference. In a conditional approach, factors ϑ_{kt} specifying the mixture component associated with the observation ξ_{kt} are generally not distinct. Therefore the distribution $F(\vartheta_{kt})$ of ξ_{kt} given ϑ_{kt} and a prior distribution G_k associated with group k for factors $\vartheta_k = (\vartheta_{k1}, \vartheta_{k2}, ...)$ are used to establish a direct link by means of k mixture components. This is

$$\begin{aligned}
&\vartheta_{kt}|G_k \sim G_k \quad \text{for each } k \text{ and } t, \\
&\xi_{kt}|\vartheta_{kt} \sim F(\vartheta_{kt}) \quad \text{for each } k \text{ and } t.
\end{aligned}$$

In a „richer get richer" environment, G is marginalized out. Consequently, a condensed clustering of the mixing parameters $\boldsymbol{\vartheta}$ is expected. A good starting point to combine both approaches parsimoniously is the Pólya Urn representation. It draws the conditional prior distribution iteratively by [15]

$$\boldsymbol{\vartheta}_t|\boldsymbol{\vartheta}_{t-1}, ..., \boldsymbol{\vartheta}_1 \sim \frac{1}{t-1+\alpha}\left(\sum_{j=1}^{t-1} \delta_{\boldsymbol{\vartheta}_j} + \alpha G_0\right), \tag{2.11}$$

$1 \leq t \leq T$, and $\delta_\bullet$ is the Dirac measure. In general, if the number of clusters is large enough, marginal- and conditional-based inferences would predicate similar clustering behavior.

2.2.2 Normal DPMMs

Let $\boldsymbol{\vartheta} = \{\boldsymbol{\vartheta}_1, ..., \boldsymbol{\vartheta}_T\}$ and $\boldsymbol{\vartheta}^{(t)} = \{\boldsymbol{\vartheta}_1, ..., \boldsymbol{\vartheta}_{t-1}, \boldsymbol{\vartheta}_{t+1}, ..., \boldsymbol{\vartheta}_T\}$. According to equation (2.11), the conditional prior distribution for $(\boldsymbol{\vartheta}_t|\boldsymbol{\vartheta}^{(t)})$ is given by

$$(\boldsymbol{\vartheta}_t|\boldsymbol{\vartheta}^{(t)}) \sim \frac{1}{\alpha+T-1}\sum_{i=1, i\neq t}^{T} \delta_{\boldsymbol{\vartheta}_i}(\boldsymbol{\vartheta}_t) + \alpha\frac{1}{\alpha+T-1}G_0(\boldsymbol{\vartheta}_t), \tag{2.12}$$

where G_0 is the prior expectation of G, i.e., $\mathbb{E}\{G(\boldsymbol{\vartheta}_t)\} = G_0(\boldsymbol{\vartheta}_t)\ \forall \boldsymbol{\vartheta}_t \in \mathbb{R} \times \mathbb{R}^+$. It follows

$$(\boldsymbol{\vartheta}_{T+1}|\boldsymbol{\vartheta}) \sim \frac{1}{\alpha+T}\sum_{t=1}^{T} \delta_{\boldsymbol{\vartheta}_t}(\boldsymbol{\vartheta}_{T+1}) + \alpha\frac{1}{\alpha+T}G_0(\boldsymbol{\vartheta}_{T+1}). \tag{2.13}$$

This means that the next case $\boldsymbol{\vartheta}_{T+1}$ is a new distinct value with probability $\alpha\frac{1}{\alpha+T}$. If $\kappa < T$ denotes the number of distinct values $\boldsymbol{\vartheta}^*_j = (\mu^*_j, \sigma^{*2}_j)$ with each t_j occurrences out of an index set

[15]Find details on the Pólya Urn representation in Blackwell and MacQueen (1973 [37])

$I_j,\ 1 \leq j \leq \kappa$, then $\boldsymbol{\vartheta}_t = \boldsymbol{\vartheta}_j^* \ \forall t \in I_j$ and $\sum_{j=1}^{\kappa} t_j = T$. Equation (2.13) can therefore be rewritten as

$$(\boldsymbol{\vartheta}_{T+1}|\boldsymbol{\vartheta}) \sim \frac{1}{\alpha+T}\sum_{j=1}^{\kappa} t_j \delta_{\boldsymbol{\vartheta}_j^*}(\boldsymbol{\vartheta}_{T+1}) + \alpha\frac{1}{\alpha+T}G_0(\boldsymbol{\vartheta}_{T+1}). \tag{2.14}$$

The amount of stochasticity in the prior distribution for κ increase monotonously with α. Escobar and West (1995 [19]) give the approximation $\mathbb{E}(m|\alpha, T) \approx \alpha \ln(1 + \frac{T}{\alpha})$ for moderately large T. To specify the prior mean G_0 of G, an appropriate form is the normal inverse gamma conjugate $(\mu, \sigma^2) \sim \mathcal{N}i\mathcal{G}(s/2, S/2, m, \tau) \sim p(\sigma^2)p(\mu|\sigma^2)$, where $p(\sigma^2) \sim i\mathcal{G}(s/2, S/2)$ and $p(\mu|\sigma^2) \sim \mathcal{N}(m, \tau\sigma^2)$. Braun (2023 [17]) uses the prior parameters $s = 10$, $S = 6/5$, $m = 0$ and $\tau = 2$ for the global crude oil model.

2.2.3 Prediction

Because the distribution of the shocks ξ_t is conditional on the parameter $\boldsymbol{\vartheta}$, predicting the random normal shock Ξ_{T+1} simplifies determining $P(\Xi_{T+1}|\boldsymbol{\vartheta}, \mathcal{D}_T) \equiv P(\Xi_{T+1}|\boldsymbol{\vartheta})$. Computationally and by means of equation (2.10), it is equivalent to evaluating $\int P(\Xi_{T+1}|\boldsymbol{\vartheta}_{T+1})\, dP(\boldsymbol{\vartheta}_{\boldsymbol{T+1}}|\boldsymbol{\vartheta})$, and using equation (2.13), one glimpses the predictive conditional density

$$(\Xi_{T+1}|\boldsymbol{\vartheta}) \sim \frac{1}{\alpha+T}\sum_{t=1}^{T} \phi(\mu_t, \sigma_t^2) + \alpha\frac{1}{\alpha+T}T_s(m, M), \tag{2.15}$$

where $\phi(\mu_t, \sigma_t^2)$ is the density of the normal distribution with mean μ_t and variance σ_t^2, and $T_s(m, M)$ the density of the Student-t distribution with s degrees of freedom, mode m and scale factor $M^{1/2}$, given $M = (1+\tau)S/s$. The extension of this result based on distinct samples of $\boldsymbol{\vartheta}_t = \boldsymbol{\vartheta}_j^* \ \forall t \in I_j$, is derivable from equation (2.14) as

$$(\Xi_{T+1}|\boldsymbol{\vartheta}) \sim \frac{1}{\alpha+T}\sum_{j=1}^{\kappa} t_j \phi(\mu_t^*, \sigma_t^{*2}) + \alpha\frac{1}{\alpha+T}T_s(m, M). \tag{2.16}$$

This result is strongly related to standard Gaussian kernel density estimates $P(\Xi_{T+1}|\mathcal{D}_n) \propto \sum_{t=1}^{T} \phi(\xi_{T+1}, \mathrm{H})$ for some global smoothing parameter H. In this case, however, the cluster strength is managed by the global smoothing parameter α, while the $\xi_t{}'$ s are regressed on their respective means μ_t, and the density estimate shrunk toward the initial prior t- distribution.

Chapter 3

Integration of DPMMs in SVARs

3.1 SVAR-DPMM methodology

To incorporate the DPMM into the multivariate SVAR model with the deterministic term η (which is the same in the reduced-form VAR), define $x_t := (y^{'}_{t-1}, ..., y^{'}_{t-p}, 1)^{'}$ and $A_+ := (A_1, ..., A_p, \eta)^{'}$. It is $y_t = \eta + \sum_{j=1}^{p} A_j y_{t-j} + u_t$, the full structural model is condensed in

$$B_0(\underbrace{y_t - A_+ x_t}_{u_t}) = \xi_t,$$

and for $1 \leq k \leq K$ each structural shock is specified by individual DPMMs hierarchically:

$$\begin{aligned} \xi_{kt}|\vartheta_{kt} &\sim F(\vartheta_{kt}), \\ \vartheta_{kt} &\sim G_k, \\ G_k &\sim DP(G_{0k}, \alpha_k), \\ G_{0k} &\sim \mathcal{N}i\mathcal{G}(s_k/2, S_k/2, m_k, \tau_k). \end{aligned}$$

As before, $F(\vartheta_{kt})$ indicates a univariate normal distribution with mean μ_{kt}, variance σ^2_{kt}, and the prior expectation G_{0k} of each G_k is chosen to be a conjugate normal inverse gamma distribution. The prior for the structural parameters in B_0 can be chosen very flexibly, as long as the different rows in B_0 remain mutually independent, and the priors are not against economic common sense. The α_k's $\in \mathbb{R}^+$ can be configured either fixed or random. In the latter case, a gamma distribution is a good choice. Similarly, priors for the m_k's and the τ_k can be chosen to follow, respectively, a normal and an inverse gamma distribution. The vectorized reduced-form slope parameter vec(A_+) can take the Gaussian form to facilitate direct inference. In the DPMM setup, note finally that marginals must not have mean zero as in Gaussian SVARs. This is not a big issue, as the intercept does not have implications for IRFs and FEVDs.

3.2 General MCMC algorithm

The construct of the MCMC algorithm is iteratively drawn from the VAR and DPMM parameters, while the central point of interest is to draw a consistent estimate for the matrix B_0. To infer the conditional posterior of B_0, the set of conditional distributions of $\alpha_+ := \text{vec}(A_+)$, of each row $B_{k\bullet}$ of B_0, and of the DPMM parameters are updated in each iteration. To easily assess each row in B_0, define n_k as the number of free elements in $B_{k\bullet}$, and write each row $B'_{k\bullet} = w_k + W_k b_k$ as a linear combination of a $n_k \times 1$ vector b_k that contains the quantities to be estimated and a $K \times 1$ vector w_k composed of restricted values and n_k zeros. W_k is a selection matrix $K \times n_k$ with zeros and ones. The b_k's are given a uniform prior. In summary, the following configurations are observed for $k = 1, ..., K$ and $t = 1, .., T$:

$$\begin{aligned}
B_0(y_t - A_+ x_t) &= \xi_t \\
\xi_{kt}|\vartheta_{kt} &\sim \mathcal{N}(\mu_{kt}, \sigma^2_{kt}) \\
b_k &\sim \text{Unif}(\{b_{ik}, 1 \le i \le n_k\}) \propto \eta \\
\alpha_+ &\sim \mathcal{N}(\mu_{\alpha_+,\Sigma_{\alpha_+}}) \\
\vartheta_{kt} &\sim G_k, \\
G_k &\sim \text{DP}(G_{0k}, \alpha_k), \\
G_{0k} &\sim \mathcal{N}i\mathcal{G}(s_k/2, S_k/2, m_k, \tau_k),
\end{aligned}$$

with potentially varying hyperpriors

$$\begin{aligned}
\alpha_k &\sim \mathcal{G}(a_{\alpha_k}, b_{\alpha_k}), \\
\tau_k &\sim i\mathcal{G}(a_{\tau_k}, b_{\tau_k}), \\
m_k &\sim \mathcal{N}(m_{m_k}, V_{m_k}),
\end{aligned}$$

in the sense that they can be considered fixed or changing along the algorithm. In sum, the strategy outlined in the last part of section 2.1.4 is executed iteratively with respect to the posterior distribution of $(\boldsymbol{\beta}|\Xi, \boldsymbol{\vartheta}) \sim (\boldsymbol{\beta}|\Xi)$, setting $\boldsymbol{\beta} := \{\alpha_+, b_k, \alpha_k, \tau_k, m_k, 1 \le k \le K\} \bigcup \{\vartheta_{kt}, 1 \le k \le K,\ 1 \le t \le T\}$ and reexamining $g_{\boldsymbol{\beta}}(\vartheta_{kt}) = g(\vartheta_{kt}|\boldsymbol{\beta})$ in each step. Here are the steps:

<u>**Step 1**</u>: For $1 \le t \le T$, split the mixing parameters into two components such that $\vartheta_{kt} = \vartheta^*_{k,\eta_{kt}}$, where η_{kt} are latent discrete variables attributing unique cluster parameters ϑ^*_{kj} to each $1 \le j \le \kappa_k := \#\{\eta_{kj}, j \ne t$ such that η_{kj} is unique$\}$, and draw the effective cluster parameters from

$(\vartheta^*_{kj}|\Xi, \boldsymbol{\beta}^{(\vartheta^*_{kj})})$, $1 \leq k \leq K$, as [1]

$$\sigma^{*2}_{kj} \sim i\mathcal{G}\left(\frac{s_k + \sum_{t=1}^T \mathbb{1}\{\eta_{kt} = j\}}{2}, \frac{1}{2}\left(S_k + \frac{m_k^2}{\tau_k} + \sum_{t=1}^T \xi^2_{kt}\, \mathbb{1}\{\eta_{kt} = j\} - \frac{\overline{m}^2_{kj}}{\overline{V}_{kj}}\right)\right), \text{ and}$$

$$\mu^*_{kj} \sim \mathcal{N}\left(\overline{m}_{kj}, \sigma^{*2}_{kj}\overline{V}_{kj}\right),$$

where $\overline{V}_{kj} = \left(\frac{1}{\tau_k} + \sum_{t=1}^T \mathbb{1}\{\eta_{kt} = j\}\right)^{-1}$, and $\overline{m}_{kj} = \overline{V}_{kj}\left(\frac{m_k}{\tau_k} + \sum_{t=1}^T \xi_{kt}\, \mathbb{1}\{\eta_{kt} = j\}\right)$.

Step 2: For $1 \leq k \leq K$, sample the hyperparameters from the respective conditional distributions.

- To α_k, consider a intermediate variable $(d_k|\alpha_k) \sim \text{Beta}(\alpha_k + 1, T)$, and draw accordingly

$$(\alpha_k|\Xi, \boldsymbol{\beta}^{(\alpha_k)}, d_k) \sim \pi_{d_k}\mathcal{G}\left(a_{\alpha_k} + \kappa_k, b_{\alpha_k} - \log(d_k)\right) \ + \ (1-\pi_{d_k})\mathcal{G}\left(a_{\alpha_k} + \kappa_k - 1, b_{\alpha_k} - \log(d_k)\right),$$

  where the odds for d_k is defined as $\frac{\pi_{d_k}}{1-\pi_{d_k}} = \frac{a_{\alpha_k}+\kappa_k-1}{T(b_{\alpha_k}-\log(d_k))}$;

- To m_k, it is

$$(m_k|\Xi, \boldsymbol{\beta}^{(m_k)}) \sim \mathcal{N}\left(m_{m_k} - \frac{V_{m_k}}{1 + \frac{\tau_k}{m_{m_k}\sum_{j=1}^{\kappa_k}\sigma^{*-2}_{kj}}} + \frac{V_{m_k}\sum_{j=1}^{\kappa_k}\sigma^{*-2}_{kj}\mu^*_{kj}}{\tau_k + m_{m_k}\sum_{j=1}^{\kappa_k}\sigma^{*-2}_{kj}}, \right.$$
$$\left. \frac{V_{m_k}}{1 + \frac{1}{\tau_k}(m_{m_k}\sum_{j=1}^{\kappa_k}\sigma^{*-2}_{kj})}\right);$$

- To τ_k, it is

$$(\tau_k|\Xi, \boldsymbol{\beta}^{(\tau_k)}) \sim \mathcal{G}\left(a_{\tau_k} + \frac{\kappa_k}{2}, b_{\tau_k} + \frac{\sum_{j=1}^{\kappa_k}\frac{1}{\sigma^{*2}_{kj}}(\mu^*_{kj} - m_k)}{2}\right).$$

Step 3: Define the block matrix $U = [u_1 \cdots u_T]'$, where $u_t = y_t - A_+ x_t$, and using a uniform prior for the b_k's, draw the conditional posterior according to an independent Metropolis Hastings step:

$$(b_k|\Xi, \boldsymbol{\beta}^{(b_k)}) \propto |B_0|^T \exp\left(-\frac{T}{2}(b_k - \mu_{b_k})'\Upsilon^{-1}_{b_k}(b_k - \mu_{b_k})\right),$$

where $\Upsilon^{-1}_{b_k} = T^{-1}W_k^{'}U\Sigma_k^{-1}UW_k$ with $\Sigma_k = \text{diag}(\sigma^2_{k1}, ..., \sigma^2_{kT})$,
and $\mu_{b_k} = (W_k^{'}U^{'}\Sigma_k^{-1}UW_k)^{-1}W_i^{'}\Xi'(\mu_k - Uw_i)$ with $\mu_k = (\mu_{k1}, ..., \mu_{kT})^{'}$.

[1] Splitting the mixing parameters into two components involve sampling conditional η_{ij}'s on one hand based on the base distribution G_0, and on the other hand based on the posterior distribution of $(\boldsymbol{\vartheta}_{k,\eta_{kj}}|G_0, \{\xi_{kj}, j \neq t\})$. Details are described in the third algorithm of Neal (2000 [23] p. 255).

Step 4: Lastly, draw from the (multivariate) conditional posterior of the VAR parameters by

$$(\alpha_+|\Xi, \boldsymbol{\beta}^{(\alpha_+)}) \sim \mathcal{N}(\overline{\mu}_{\alpha_+}, \overline{V}_{\alpha_+}),$$

where $\overline{V}_{\alpha_+} = \left(V_{\alpha_+}^{-1} + \sum_{t=1}^{T}(x_t \otimes I_K)(B_0^{'}\Sigma_t^{-1}B_0)(x_t^{'} \otimes I_K)\right)^{-1}$ with $\Sigma_t = \text{diag}(\sigma_{1t}^2, ..., \sigma_{Kt}^2)$ and $\overline{\mu}_{\alpha_+} = \overline{V}_{\alpha_+}\left(V_{\alpha_+}^{-1}m_{\alpha_+} + \sum_{t=1}^{T}(x_t \otimes I_K)(B_0^{'}\Sigma_t^{-1}B_0)(y_t - B_0^{-1}\mu_t)\right)$ with $\mu_t = (\mu_{1t}^2, ..., \mu_{Kt}^2)^{'}$.

3.3 Posterior inference

Posterior analysis is commonly pursued with the help of IRFs and FEVDs. The variance and historical decompositions are derived from the unconditional variance of the structural shocks obtained from the MCMC related predictive density. Since the posterior for B_0 is

$$P(B_0|\Xi) \propto \underbrace{P(\Xi|B_0)}_{\text{likelihood}} \quad \underbrace{P(B_0)}_{\text{prior}},$$

where integrating out $\boldsymbol{\vartheta}$ provide the likelihood

$$P(\Xi|B_0) = \int P(\Xi|\boldsymbol{\vartheta}, B_0)P(\boldsymbol{\vartheta}|B_0)\ d(\boldsymbol{\vartheta}),$$

the Bayesian density estimation is fully solved by applying equations (2.15) or (2.16) to draw the unconditional predictive distribution

$$P(\Xi_{T+1}|B_0, u_t) = \int P(\Xi_{T+1}|\boldsymbol{\vartheta})\ dP(\boldsymbol{\vartheta}|B_0, u_t).$$

Based on the posterior distribution for B_0, the MCMC algorithm basically generates a list of multiple possible choices, say $\tilde{B}_0$, for which the (log) likelihood is calculated with respect to mutually shock independency:

$$\ell(\tilde{B}_0) = log \prod_{t=1}^{T} \widehat{f}(\xi_t|\tilde{B}, u_t) = \sum_{t=1}^{T} log(\underbrace{\widehat{f}(\xi_t|\tilde{B}_0, u_t)}_{\prod_{k=1}^{K}\widehat{f_k}(\xi_{kt}|\tilde{B}_0, u_t)}),$$

where the multivariate joint densities are generated by means of the DPMMs for ξ_{kt} described above. Finally, maximizing the logarithmic likelihood $\ell(\tilde{B}_0)$ gives the best expectation

$$\widehat{\tilde{B}}_0 = \underset{\tilde{B}_0}{\text{argmax}}\ \ell(\tilde{B}_0).$$

3.4 Convergence properties

3.4.1 Strong identification by non-Gaussianity

Consider synthetic data sample of size $T = 500$ from the small introductory example in section 2.1.1 (stylized bivariate model of supply and demand), configured with $\gamma = 0.05$, $\lambda = -0.35$, $\sigma_1 = 1$, $\sigma_2 = 0.5$, and standardized shocks $\xi_t^\bullet = \sqrt{\frac{\nu}{\nu-2}}\tilde{\xi}_t^\bullet$. The setting $\tilde{\xi}_t^\bullet \sim t_3$ corresponds to strong identification from non-Gaussianity, since the student-t distribution with $\nu = 3$ degrees of freedom has heavier tails than a Gaussian distribution. The model is estimated with the priors $\gamma \sim T_3(0.1, 0.2^2)$ and $\lambda \sim T_3(-0.1, 0.2^2)$ without sign restrictions. Generating 1000 draws from the MCMC algorithm takes approximately $1,07$ seconds from a $i5$ laptop Intel (R) Core (TM) $i5 - 1235$ CPU with 1.30 GHz. Figure 3.1 shows the structural shocks, respectively, for supply and demand along with the posterior predictive density and 90% credible sets estimated by the SVAR-DPMM model, which obviously can capture the amount of non-Gaussianity in the data. Thus, the second shock, for example, has strong outliers that produce very heavy tails.

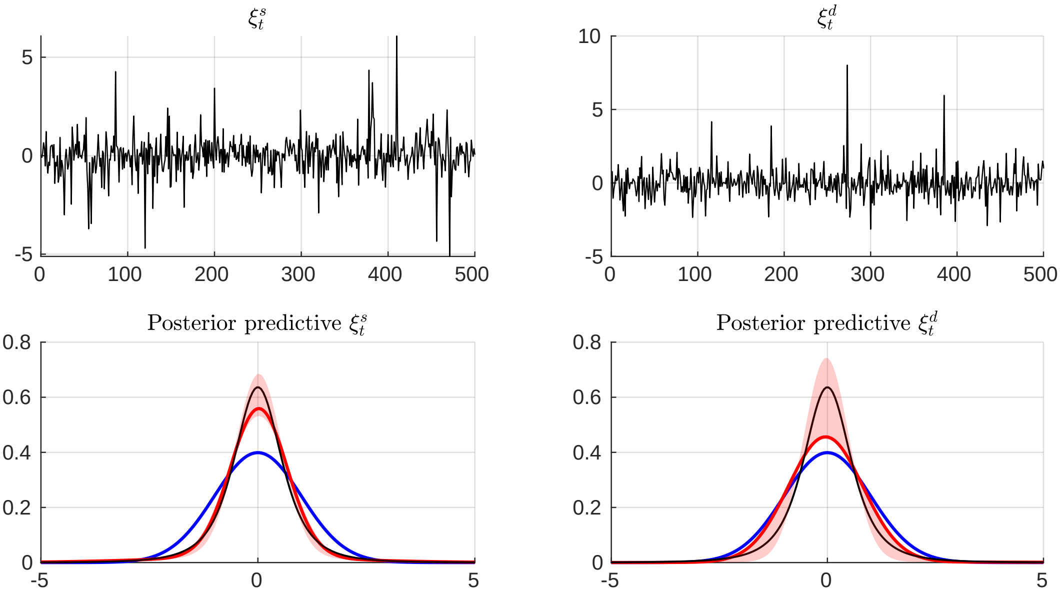

Figure 3.1: Strong identification by non-Gaussianity, with $\nu = 3$. In the upper panel are simulated structural shocks. Below posterior predictive densities in the non-Gaussian setup (red lines). The red areas are 90% credible sets, the black lines a unit-variance standardized t_3 density, and the blue line gives $\phi_{(0,1)}$.

Next, a Markov chain of length 10000 for the two structural parameters generated by saving every 10th draw is observed in the top panel of figure 3.2. Visual inspection suggests that the

MCMC has converged well. However, the tendency proposes that a larger Markov chain would lead to much more precise results, probably resembling those obtained by generating $i.i.d.$ draws. The comparison between priors and posteriors shows that the data are fairly informative, as the posterior mass narrows the value of γ and moves towards the true value λ.

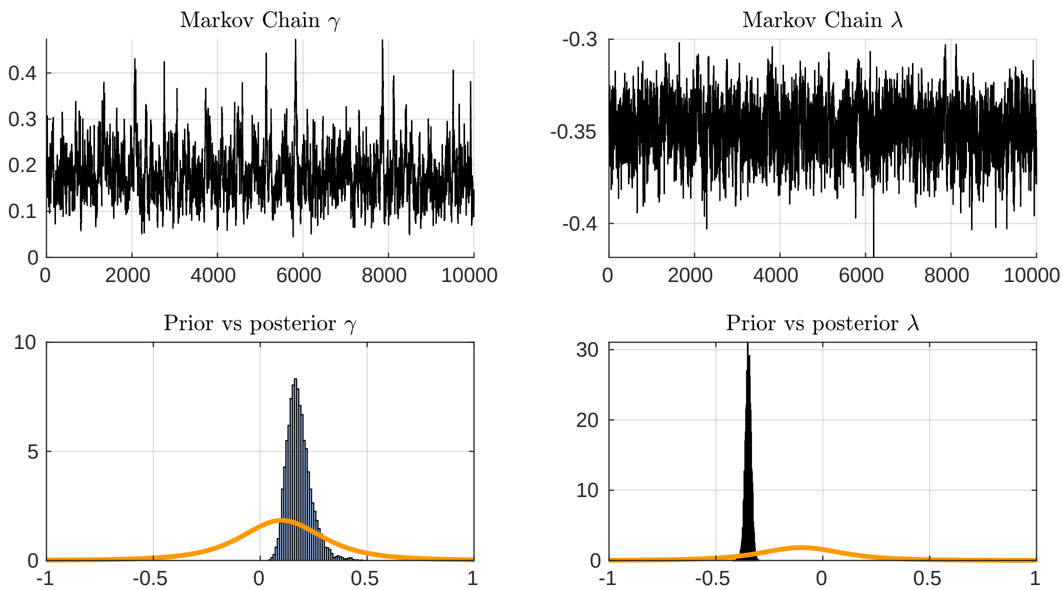

Figure 3.2: Strong identification by non-Gaussianity, with $\nu = 3$. Top panel: MCMC output of length 10000 in the non-Gaussian setup. Bottom panel: Prior (orange line) and posterior of the two structural parameters γ and λ.

3.4.2 Weak identification by non-Gaussianity

Figures 3.3 and 3.4 are the same convergence representations, however, with a higher degree of freedom $\nu = 10$, i.e. where standardized shocks yield considerably less identification information from non-Gaussianity (weak identification), the simulated shocks are closer to normality. Thus, the 90% credible set covers the Gaussian bell curve largely. However, MCMC does not seem to have been efficient. Even with a larger Markov chain, convergence is not assured. Furthermore, the posterior is less informative about γ, whereas somewhat remain informative for λ, however, for an MCMC which has not converged.

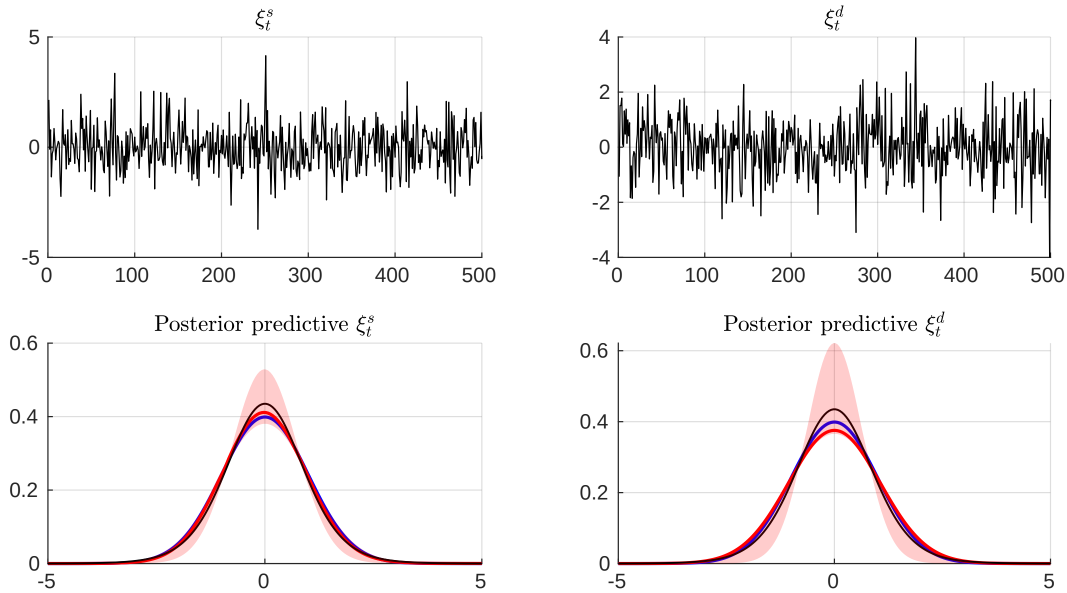

Figure 3.3: Weak identification by non-Gaussianity, with $\nu = 10$. Top panel: Structural shocks. Bottom panel: posterior predictive densities from the SVAR-DPMM along with 90% credible sets.

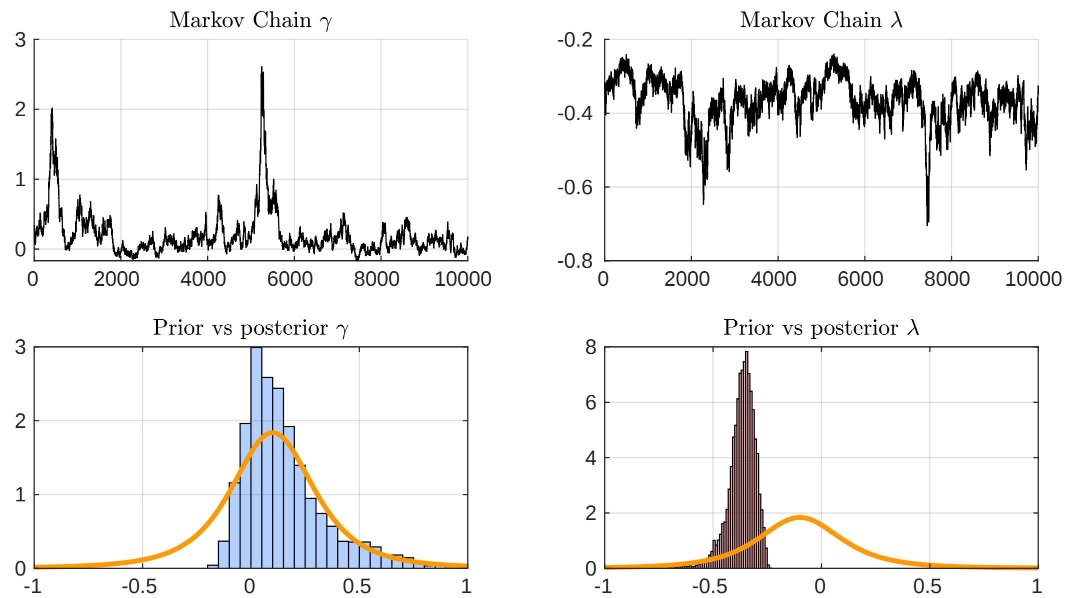

Figure 3.4: Weak identification by non-Gaussianity, with $\nu = 10$. Top panel: Markov chains for the structural parameters γ and λ. Bottom panel: Priors (orange lines) VS posteriors.

Chapter 4

Empirical application: The importance of supply and demand for oil prices

4.1 Structural shock estimation

Through the empirical part, the theoretical results are applied to the following four variable oil market model examined in Kilian and Murphy (2014 [12]), Baumeister and Hamilton (2019 [15]) and most recently Braun (2023 [17]):

$$y_t = (100 \times \Delta q_t, 100 \times \Delta y_t^a, 100 \times \Delta p_t, \Delta i_t)^{'}.$$

The data are monthly. Four variables are observed during the period from January 1958 until December 2022: the log of global crude oil production in million barrels per day (q_t), a proxy for oil demand related to the industrial production of the OECD and six additional countries (y_t^a), the log of the real oil price (p_t), and a proxy for OECD oil inventories expressed as a fraction of global crude oil production (Δi_t). The data from January 1958 until December 1973 is used only as a training set for an eventual prior settlement. Five uncorrelated structural shocks with regard to supply (ξ_t^s), economic activity (ξ_t^{ad}), consumption demand (ξ_t^{cd}), inventory demand (ξ_t^{id}) and measurement error (ξ_t^{me}) must be recovered from the available samples. This is $\xi_t = (\xi_t^s, \xi_t^{ea}, \xi_t^{cd}, \xi_t^{id}, \xi_t^{me})^{'} \sim (0, \Sigma_\xi)$ with $\Sigma_\xi = \text{diag}(\sigma_1^2, ..., \sigma_5^2)$. Further, following Braun (2023), the structural oil market is delineated according to five simultaneous equations:

$$\begin{aligned}
&\textit{Supply:} \quad q_t = \gamma_{qp} p_t + \xi_t^s, && (4.1)\\
&\textit{Economic activity:} \quad y_t^a = \gamma_{yp} p_t + \xi_t^{ad}, && (4.2)\\
&\textit{Consumption demand:} \quad q_t - i_t^\star = \lambda_{qy} y_t^a + \lambda_{qp} p_t + \xi_t^{cd}, && (4.3)\\
&\textit{Inventory demand:} \quad i_t^\star = \psi_1 q_t + \psi_3 p_t + \xi_t^{id}, && (4.4)\\
&\textit{Measurement error:} \quad i_t = \chi i_t^\star + \xi_t^{me}, && (4.5)
\end{aligned}$$

where γ_{qp} stands for the oil price elasticity of oil supply, γ_{yp} for the price effect on industrial production, λ_{qy} for the income elasticity of oil demand, λ_{qp} for the price elasticity of oil demand, ψ_1 and ψ_3 respectively for quantity and price effects on inventory demand, and χ for the OECD fraction of true oil inventories (about $60-65\%$). In matrix form, the structural model corresponding to the system of equations (4.1), (4.2), (4.3), (4.4), and (4.5), is given by

$$\underbrace{\begin{pmatrix} \xi_t^s \\ \xi_t^{ea} \\ \xi_t^{cd} \\ \xi_t^{id} \\ \xi_t^{me} \end{pmatrix}}_{\text{structural shocks } \xi_t} = \underbrace{\begin{bmatrix} 1 & 0 & -\gamma_{qp} & 0 & 0 \\ 0 & 1 & -\gamma_{yq} & 0 & 0 \\ 1 & -\lambda_{qy} & -\lambda_{qp} & 0 & -1 \\ -\psi_1 & 0 & -\psi_3 & 0 & 1 \\ 0 & 0 & 0 & 1 & \chi \end{bmatrix}}_{\text{structural matrix } B_0} \underbrace{\begin{pmatrix} u_t^q \\ u_t^{y^a} \\ u_t^p \\ u_t^i \\ u_t^{i^\star} \end{pmatrix}}_{\text{reduced-form VAR errors } u_t}. \tag{4.6}$$

4.2 Adjustment for the oil market model

The number of lags is chosen to be $p = 12$, so that enough dynamics are taken into account. Given that $u_t^{i^\star}$ is unobserved and latent with respect to global inventory series, the model in equation (4.6) cannot be identified from second moments of the data.[1] Two models are considered that are different only by the specification of the error term. The first Gaussian model uses a Gaussian likelihood for all shocks, in particular $\xi_t \sim \mathcal{N}(0, \Sigma_\xi)$, where each $\sigma_k, k = 1, ..., 4$ are given weekly informative inverse gamma priors. To express prior expectations on σ_5 with orthogonal shocks, the nonlinear transformation $\rho = \frac{\chi^{-1}\sigma_5^2}{\sigma_3^2 + \chi^{-2}\sigma_5^2}$, is used, $0 \leq \rho \leq \chi$, and given therefore a Beta prior to an uncritical extend, i.e. centered around 0.25χ. With respect to the clauses of Manne, Meitz, and Saikkonen (2017 [25] pp. 289-291) emphasized in section 2.1.1, the second non-Gaussian model estimate all shocks but the latent measurement error by DPMMs. Preserving mutual independence among the first four shocks, it models each marginal by a DPMM of the form $\xi_{it}|\vartheta_{kt} \sim F(\vartheta_{kt})$, where F is the univariate normal distribution parameterized by mean and variance $\vartheta_{kt} = (\mu_{kt}, \sigma_{kt}^2) \sim G_k$, $G_k \sim \mathrm{DP}(G_{0k}, \gamma_k)$, $G_{0k} \sim \mathcal{N}i\mathcal{G}(s_k/2, S_k/2, m_k, \tau_k)$, $k = 1, ..., 4$, and for the measurement error, it simply holds $\xi_t^{me} \sim \mathcal{N}(0, \sigma_5^2)$. To specify a prior for σ_5 the strategy

[1] In sum, there are 12 structural parameters (i.e. 7 elements in B_0 and 5 elements in Σ_ξ), whereas there are only $4\frac{4+1}{2} = 10$ reduced-form parameters available in Σ_u, i_t and i_t^* being linearly dependent.

used in the Gaussian setup is no longer consistent. Not only would a prior on ρ instead of directly on σ_5^2 induce prior dependence between structural parameters and the DPMM of the third shock, but also the conditions of the MCMC algorithm would no longer be fulfilled. Braun (2023 [17]) proposes to break the dependence rather by posing the prior belief on the fraction of variance in the latent inventories explained by the measurement error. Since $u_t^i = \chi^{-1} u_t^{i^\star} + \xi_{5t}$, it can be inferred that the resulting coefficient is given by $\rho^\star = \frac{\sigma_5^2}{\chi^{-2}\mathrm{var}(u_t^i)+\sigma_5^2} \in (0,1)$, where $\mathrm{var}(u_t^i) \approx 1.3$ is evaluated based on the data sample from January 1958 until December 1973, and the Jacobian of transformation is computed at the stage of creating $A_{5\bullet}$ with uniform priors.[2] In this way the Jacobian can directly be used in the Metropolis-Hastings step. Lastly, a Beta prior with mean 0.25 and standard deviation 0.12 is posed on ρ^*.

For both models, the set of prior distributions for elements of B_0 are the same. Student-t distributions are assumed for all structural parameters in B_0, but the fraction of inventories χ retained by the OECD countries, of which the prior is assumed to follow a Beta distribution with mean 0.6 and standard deviation 0.1, reflecting a soft level of uncertainty. The price elasticities of supply (γ_{qp}) and demand (λ_{qp}) have student-t distributions concentrated around 0.1 and -0.1, respectively, variance 0.2 and 3 degrees of freedom. The prior for λ_{qy} is t-distributed with mode 0.7, scale 0.2 and 3 degrees of freedom. As the effect of oil prices on economic activity γ_{yp} is intended to be rather small, a t-distribution with mode -0.05 and 3 degrees of freedom is selected. Because prior knowledge of ψ_1 and ψ_3 is not available, uninformative student-t priors concentrated around 0 with scale 0.5 and 3 degrees of freedom are used.

4.3 Empirical results

4.3.1 Standardized structural shocks

To evaluate how both models rank within the scope of the world's oil market model, it is important first to make sure that both non-Gaussianity and independence assumptions on the shocks are empirically tenable. A brief look at Figure 4.1 provides evidence of non-Gaussianity for at least $\tilde{K} - 1 = 3$ shocks, where $\tilde{K}$ represents the number of shocks in the non-augmented model (without measurement error). In particular, the posterior predictive (marginal) densities of standardized shocks associated with supply, economic activity, and consumption demand along with 90% credible intervals reveal greater credibility in favor of the non-Gaussian model. Comparing the shape of the predictive estimates with the one the standard Gaussian distribution yield considerably heavier tails, especially lower variances, and a tiny mass concentration around mean zero. Overall, there is great potential to gain consistent identification information using the non-Gaussian model, based on crude oil data.

Because of the nonparametric nature of the shocks modeled by independent DPMMs,

[2]This assumption doesn't necessarily imply a uniform prior for χ and $\rho^\star$.

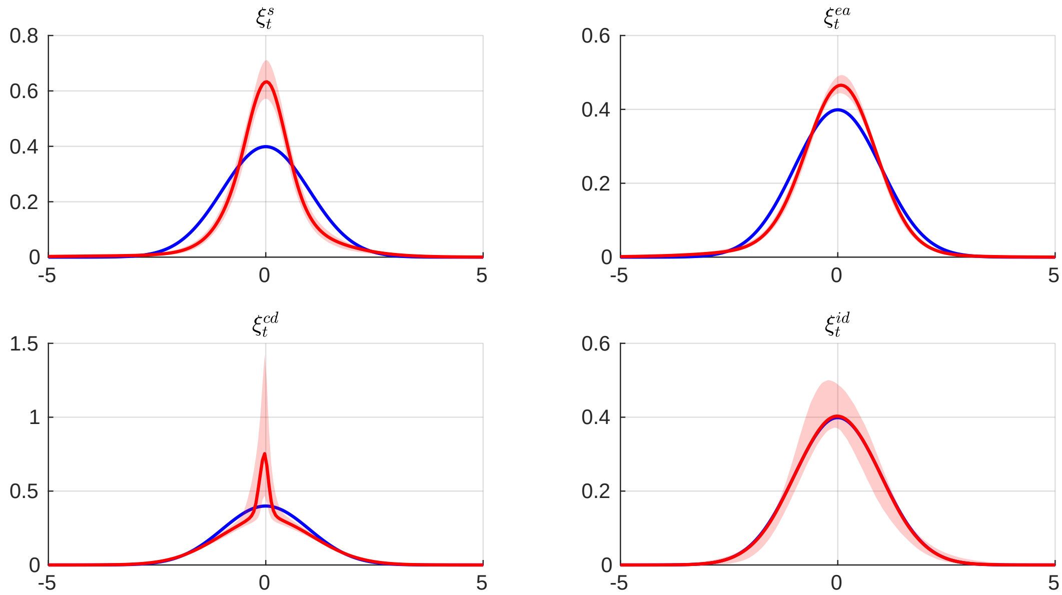

Figure 4.1: Posterior predictive marginals along with 90% credible sets. Standardized structural shocks are used. The blue lines give $\phi_{0,1}$.

standard independence test statistics such as Bayes factors may not be sufficient. From the set of all possible tests that account for non-Gaussianity, multimodality, and flexible distributions, two frequentist test statistics particularly adapted for time series SVARs are considered. The first one is nonparametric and relies on the distance covariance (DC) multivariate dependence measure defined by Matteson and Tsay (2017), which is zero only by mutual independence. It is defined by $U((\xi_1, ..., \xi_T)^{'}) = T\sum_{j=1}^{K-1} \text{DC}(\hat{U}_j, \hat{U}_{\{l|j<l\leq K\}})$, where $\hat{U}_j = \{\hat{u}_{i,k} = \frac{1}{T}\text{rank}\{\xi_{ij}\}\}$, and points to highly plausible independence in the non-Gaussian model. The amount of dependence found in the Gaussian model is rather expected, as it does not inherently account for independency. The second independence test statistic is presented in Olea, Plagborg-Møller, and Qian (2022) as $S(E) = \sqrt{\frac{1}{K(K-1)}\sum_{i=1}^{K}\sum_{i\neq j}\text{Corr}(\xi_{it}^2\xi_{jt}^2)^2}$. It is specialized for higher-order dependencies, which may not be obvious from the point of view of typical dependence shapes. For an overview, the posterior of both approaches approximated by bootstrap is provided in Figure 4.2.

4.3.2 Structural B_0 components

Prior assumptions of non-Gaussianity and independence of the structural shocks being tested, the question arises next to study how efficient was the estimation of the B_0 matrix by the MCMC algorithm adjusted for the world crude oil market model. Naturally, based on the results obtained for standardized shocks, that posterior expectations for structural parameters would differ across both setups. Comparison priors (orange lines) versus posterior distributions

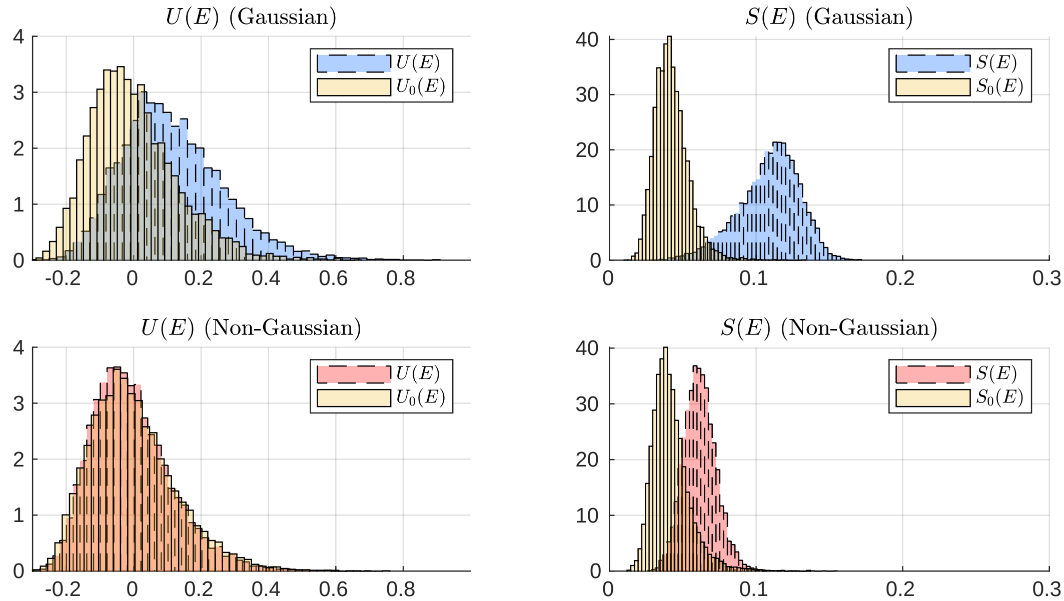

Figure 4.2: Posterior predictive distribution of the cross-referencing test statistics of Matteson and Tsay (2017) and Olea, Plagborg-Møller, and Qian (2022). Top panel: Gaussian setup. Bottom panel: Non-Gaussian setup. $U_0(E)$ and $S_0(E)$ represent the respective test statistic under the null hypothesis of mutual independent shocks.

of key structural parameters γ_{qp}, γ_{yp}, λ_{qy} and λ_{qp} in Gaussian and non-Gaussian identification strategies are shown in Figure 4.3. At first glance, the short-term price elasticities of oil supply (γ_{qp}) and of oil demand (λ_{qp}) are the most concerned by the difference of the strategy used. Although the Gaussian model tends to mirror the prior mode despite the lower uncertainty achieved, the non-Gaussian expectation strongly points to 0 for γ_{qp} and is widely revised for λ_{qp}. For the impact of the oil price on economic activity (γ_{yp}), almost the same mass concentration is expected around 0. Regarding the income elasticity of oil demand λ_{qy}, both models seem to rank the same with modal values below two. In contrast to λ_{qy}, where a very low degree of posterior mass is associated with lower values in the Gaussian setup, a much larger quantity of posterior mass is related to lower values for the short-term oil demand elasticity λ_{qp} in the non-Gaussian model, with a modal value of approximately -1.1. Figure 4.5 shows the Markov chains for the key structural parameters in the non-Gaussian MCMC approximations, which seems to have converged well. The relative numerical efficiencies (RNE) accompany each subplots title as a summary statistic of the underlying autocorrelation registered. Geweke (1992 [38] pp. 8-9) describe the RNE as the interpretation of the ratio of number of replications required to achieve the same efficiency as drawing i.i.d. from the posterior. As the RNE registered approach one when multiplying by 10, one should consider a larger Markov chain of e.g. 100000 to obtain precise results nearing those of at least 1000 i.i.d. draws.

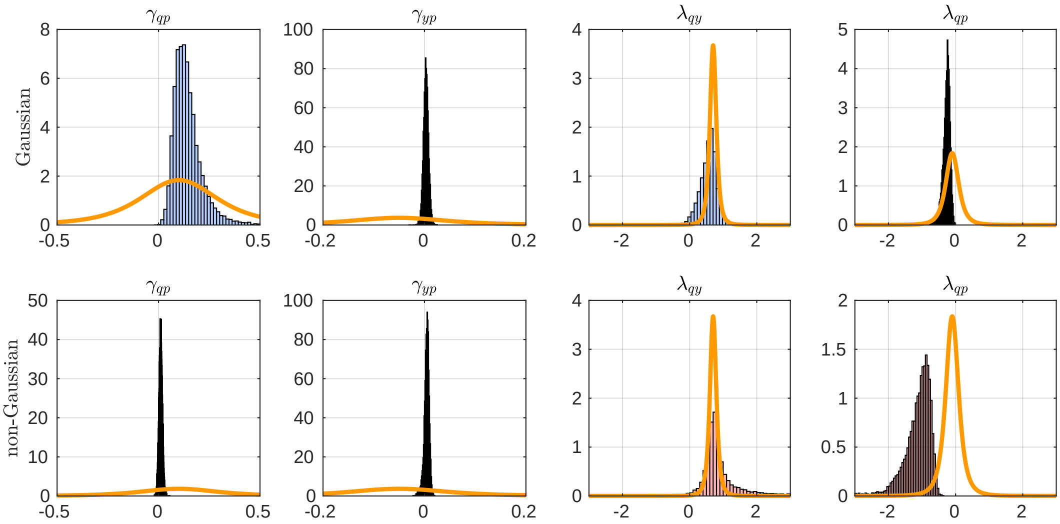

Figure 4.3: Comparison priors (orange lines) versus Posterior mass for key structural parameters.

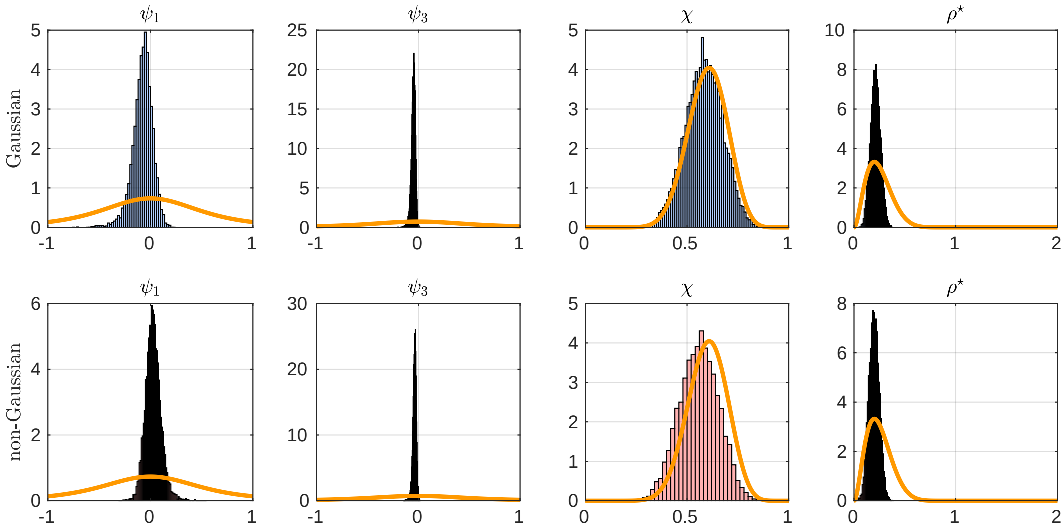

Figure 4.4: Comparison priors (orange lines) versus Posterior mass for remaining structural components.

Lastly, similar comparisons are shown in Figure 4.4 for the remaining parameters of the structural matrix B_0, and in the lower row of Figure 4.5 the Markov chain results. Gaussian and nonnormal identification schemes yield approximately the same results for common parameters. For the effect of quantity on inventories (ψ_1), the posterior mass is concentrated close to 0 with a lower volatility in the non-Gaussian setup. The impact of price on inventories (ψ_3) is clearly 0 for

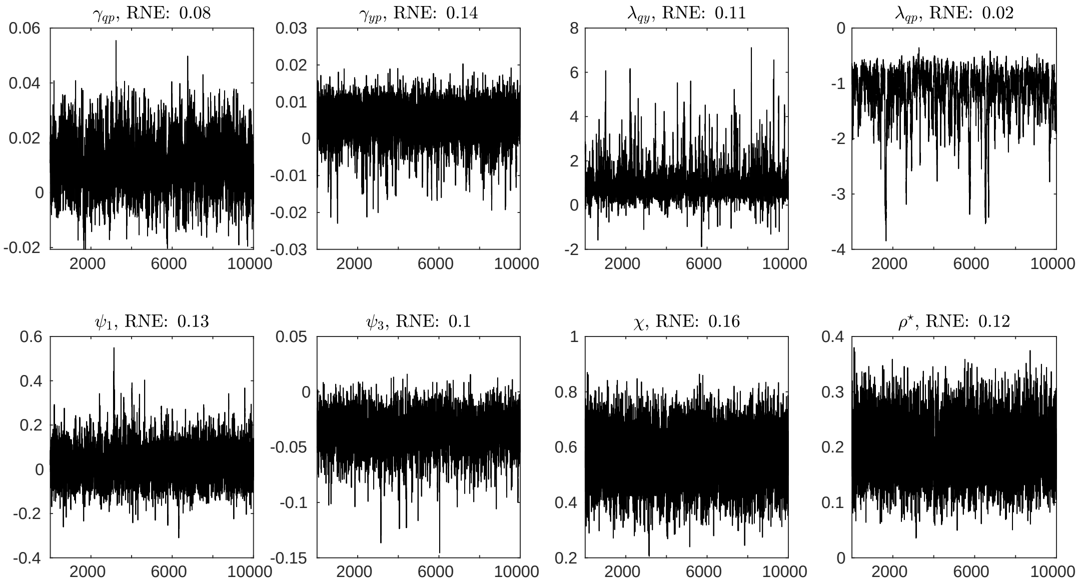

Figure 4.5: Markov chains of all structural parameters.

both perspectives, supporting the previous assumption on ψ_3. However, both the posterior of the fraction of inventories (χ) and of the importance of measurement error in u_t^{i*} (ρ^*) also seems to have converged to the prior Beta assumptions, in the case of ρ^* with much greater certainty than for χ.

4.3.3 IRFs and FEVDs

To perform the structural analysis, Figure 4.6 emphasizes differences in both identification strategies due to different posterior distributions of key structural parameters. It shows posterior medians for IRFs with 90% credible sets across 16 months, with respect to the structural dynamism within the scope of the nonaugmented model, i.e., with four endogenous variables. In the first row, it appears that much more intensity is needed for supply shocks from the non-Gaussian model to create a price augmentation than those of the Gaussian model. A slightly opposite behavior is observed in price responses for shocks related to consumption demand, although the non-Gaussian median lies above the Gaussian one for both shock types. For aggregate industrial demand shocks, the major difference for both strategies is obvious in all four columns. Shocks driven by inventory demand also tend to produce same estimates for both models, but responses to oil inventories (fourth column) are almost opposite to those of oil production (first column), however, with much wider credible sets from the non-Gaussian point of view. This may signal a lower impact of inventory demand when it comes to non-Gaussian and mutually independent innovations. In general, supply may have a greater impact on oil production and prices than demand, as IRFs generally point to increased responses for supply-driven shocks in comparison to demand-driven shocks.

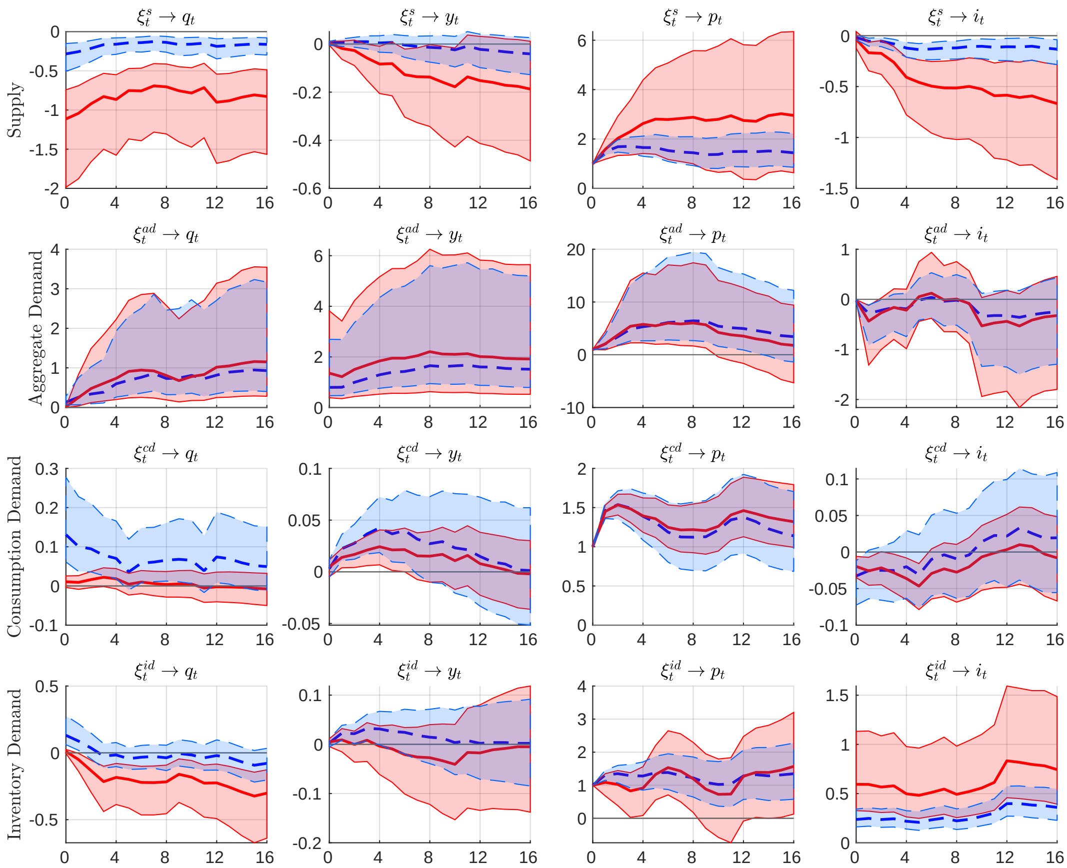

Figure 4.6: Posterior median IRFs along with 90% credible sets of the non-Gaussian model (red lines) and of the Gaussian model (blues lines).

To close this part, the FEVD of the real price of oil in the 4 and 16 months horizon with 90% credible sets in brackets, is summarized in Table 4.1. Here, also, major differences are found in both perspectives for supply-driven and consumption-demand shocks. Innovations in supply are found to be significantly larger in the Gaussian model (35% median estimate), while the 90% credible set covers the area between 15% and 65%, versus only 2% and 10% in the non-Gaussian model (5% median estimate) on the horizon 4. Similar behavior is observable at the horizon 16, where supply shocks end up allowing the total variation to be explained below 12%. The opposite trend holds for consumption demand shocks. Gaussian 90% credible sets indicate between 25% and 76% of the total variation, while non-Gaussian estimates are much larger, i.e. between 75% and 95%.

In summary, one can easily observe a inelastic oil supply and a very elastic oil consumption

Shock	ξ_t^s		ξ_t^{ea}		ξ_t^{cd}		ξ_t^{id}		ξ_t^{me}	
Horizon / Model	G	NG	G	NG	G	NG	G	NG	G	NG
4	**0.35**	**0.05**	0.05	0.03	**0.54**	**0.90**	0.03	0.01	0.01	0.01
	(0.15,0.65)	(0.02,0.10)	(0.02,0.09)	(0.01,0.07)	(0.26,0.76)	(0.83,0.95)	(0.01,0.08)	(0.00,0.02)	(0.00,0.03)	(0.00,0.03)
16	**0.32**	**0.06**	0.07	0.05	**0.50**	**0.82**	0.03	0.01	0.06	0.06
	(0.15,0.59)	(0.03,0.10)	(0.04,0.11)	(0.02,0.09)	(0.25,0.69)	(0.75,0.88)	(0.01,0.09)	(0.00,0.03)	(0.03,0.10)	(0.03,0.10)

Table 4.1: FEVD of the real price of oil.

demand in both models, while the demand elasticity is much more pronounced in the model with independent non-Gaussian shocks. Furthermore, the non-Gaussian model describes supply-driven shocks to have only a minor impact on price fluctuation, while demand-driven shocks explain a very large amount of the total variation. This findings may help to make estimates more accurate in the context of the world's crude oil market; however, more weight must be put on the identification by independence and non-Gaussianity if statistical test statistics point in favor of those assumptions.

Chapter 5

Aspects to regulations on CO_2 Emissions and natural energy sources

5.1 Using local projections

In general, policies must take into account real-world energy shocks to avoid unintended emissions. The heart of the problem lies indeed in the question of how to define smooth transition process(es) to reach net zero, i.e. equilibrium. For example, when price shocks have a strong impact on economic activity, carbon pricing may significantly reduce higher CO_2 emissions in the near term. If consumers are price-inelastic, renewable energy or policy timing for (green) energy investment subsidies may play a crucial role in the realization of lower CO_2 emissions. As the results from the last part suggest a very elastic oil demand, policies aiming to address climate change by reducing the demand for fossil fuel-based energy might have a significant impact on oil prices, but not necessarily on crude oil production. Therefore, the ultimate goal of climate change may not be achieved only by carbon pricing in the long term. To assess the potential miss-comprehension of effective instruments, it is necessary to establish the relation between GHG emissions and oil production, especially from a non-Gaussian point of view. As briefly mentioned in the second section, the best which can be aimed at is to get inside into respective local projections.

More precisely, local projections have the advantage of a very flexible application in situations where an exogenous shock is already identified from a SVAR. IN particular, once an exogenous shock has been identified, it is straight forward to estimate impulse responses using OLS regressions of the form

$$y_{t+h} = \eta^h + \tilde{\beta}_h \xi_t + \phi x_t + u^h_{t+h}, \tag{5.1}$$

$0 \leq h \leq H - 1$, where x_t is a vector of control variables, ξ_t the identified shock variable, $\tilde{\beta}_h$ the response of y at time $t + h$ to the shock variable at time t, and ϕ the regression coefficient of the control variable. For identified endogenous shocks ξ_t, the model can be adapted for a two-stage least squares regression.[1]

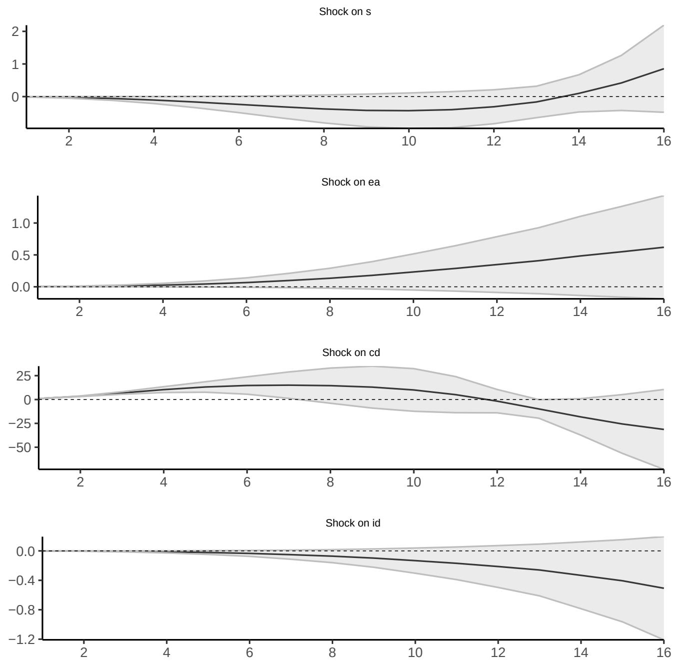

Figure 5.1: .

5.2 Interpretation of results

Figure 5.1 shows impulse responses estimated by local projections with an instrument variable that is correlated with consumption demand. The confidence bands are fixed at 90%. Local projections reveal that all shocks react almost similarly to an increase in consumption demand and oil production in the short term. However, the impacts appear to differ in the long run. At about 8 months, a clear distinction begins to occur and local projection curves indicate that economic activity has a predominant impact on CO_2 emissions and would significantly increase oil production after 16 months. The opposite behavior is expected for inventory demand, while the impact of supply on CO_2 emissions remains rather minor, unspecified.

[1]Details of the two-stage least square estimation can be found in Adämmer (2019 [39] p. 424).

Chapter 6

Conclusion

6.1 Summary

This work was concerned with the estimation and identification of independent non-Gaussian shocks in SVARS using the nonparametric framework of DPMMs. For this, a brief introductory background was recalled, motivation, topic relevancy, and objectives clarified. Then the SVAR model setup was highlighted with the nonparametric Bayesian perspective with Dirichlet processes as efficient priors to model uncertain distributions. To integrate DPMMs in their basic form with SVARs, a general MCMC algorithm was presented and its convergence properties in the case of strong and weak identification schemes. Furthermore, an empirical application was performed on the basis of acquired theoretical knowledge, and an adaptation of the previous MCMC was clarified for the world crude oil market. The empirical results led to consistent estimation of standardized structural shocks, the structural matrix B_0, IRFs, and FEVDs. To conclude, local projections were briefly performed to obtain an apercu on potential regulations to reduce the world's CO_2 emissions, and the results were briefly interpreted.

6.2 Achievements and limitations

It has been a great experience to reveal the details that constitute the application of the statistically time-series principle on a real theme, such as energy production and climate change, based on an actual data set and a model already widely studied within the scope of SVARs. The rather inelasticity of oil supply when it comes to price fluctuations, and the very strong impact of oil demand were statistically elucidated. Potential tests were also applied to confirm the identification assumption in structural shocks. Of course, the present work can be extended to more precise results using, for example, longer Markov chains, in combination with a much more efficient computer to reduce computation times, but nevertheless the goals settled at the beginning have been reached in one way or another.

6.3 Future works

Having tried to estimate the uncertain densities by parametric student-t distributions gave approximately the same results as those of DPMMs. There can be many reasons for this. Potential future work may likely consider comparing both points of view to check which one ranks as best estimation in the context of independent non-Gaussian shock estimations.

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